NEUR 411 · NEUROETHOLOGY · FALL 2026

Lecture 91

Sea lions: rhythmic entrainment, auditory–motor coordination, and experimental tests of beat synchronization

Date TBD

Jordan Lopez, Ph.D.
Professor of Neurobiology & Biomedical Engineering

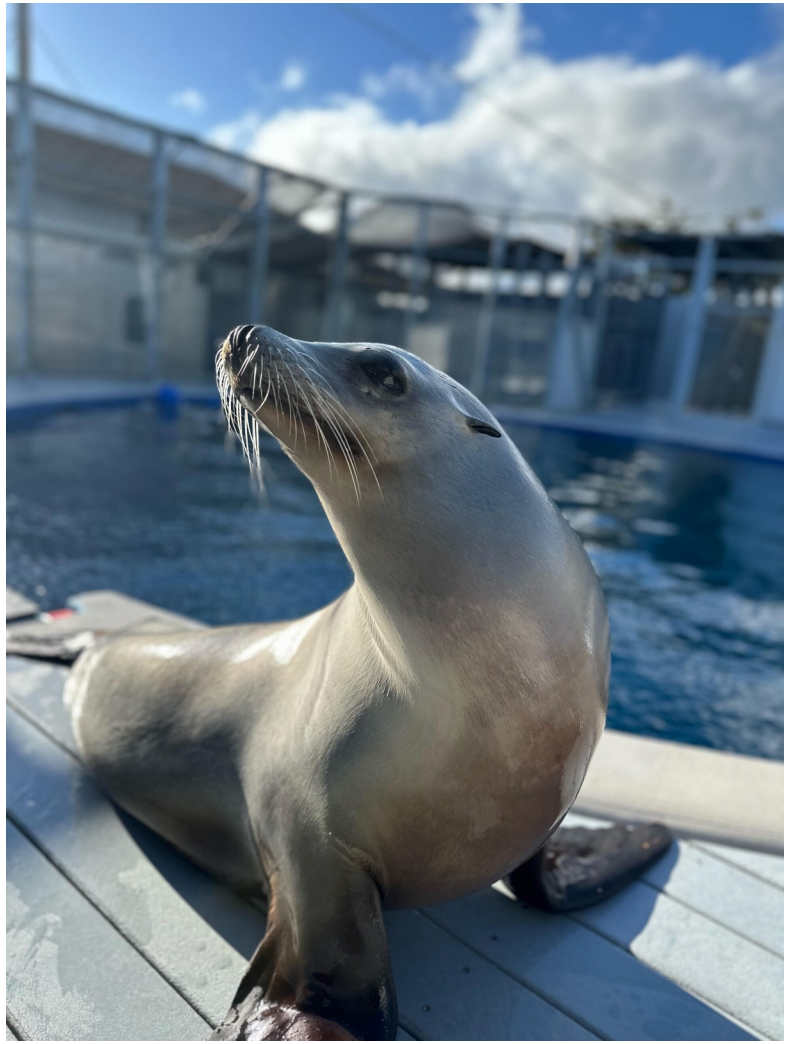


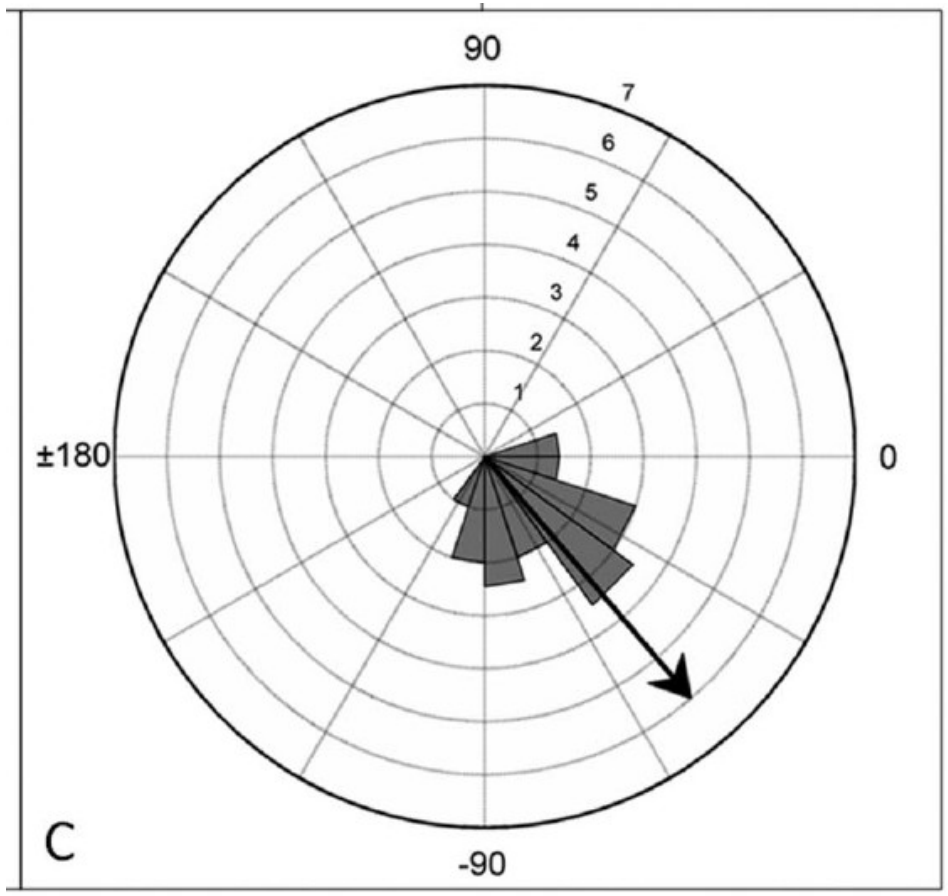
Photo: California sea lion Ronan at UC Santa Cruz’s Long Marine Laboratory. Photo by Colleen Reichmuth; NOAA/NMFS 23554.

Ronan learns to align movement with sound

Rhythmic entrainment is the adjustment of a repeating movement to an external rhythm. Ronan, a California sea lion, learned to move her head up and down while hearing regularly spaced sounds. The lowest point of each movement supplied a repeatable timing marker.

Cook and colleagues trained this behavior at 80 and 120 beats per minute, then tested other rates and musical recordings. A movement that stays aligned across unfamiliar rates requires flexible timing rather than the repetition of one memorized motor speed.

Ronan lived at the University of California, Santa Cruz. The experimental task underlying her public reputation as a dancing sea lion combined a controlled sound sequence, concealed trainers, video measurements, and food reinforcement for appropriately timed behavior.



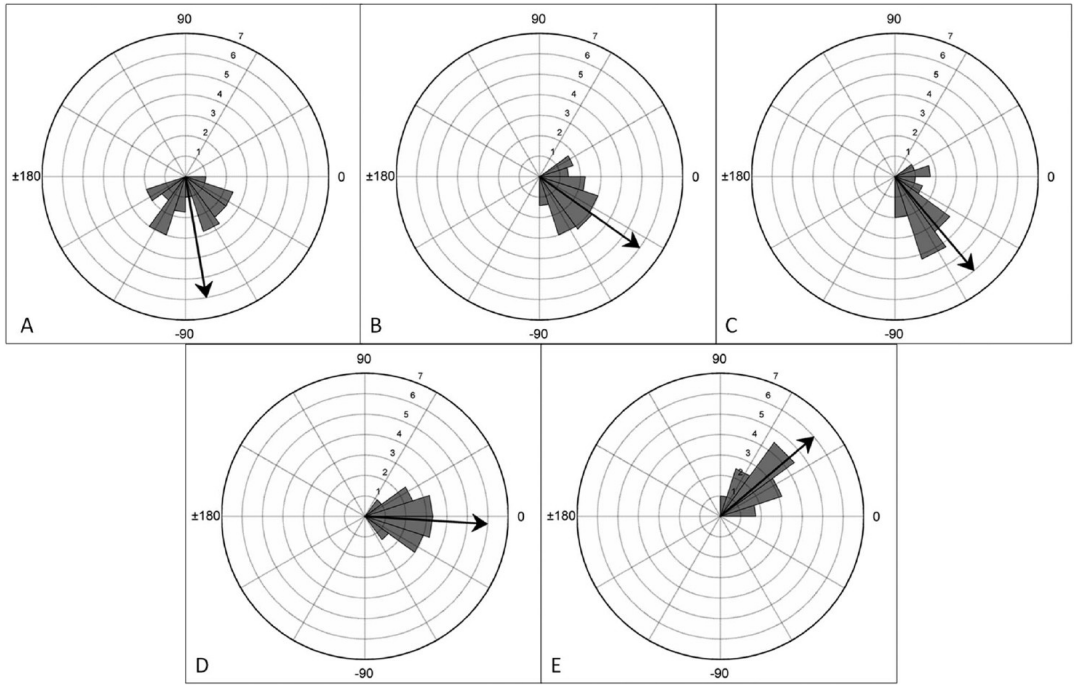
Cook et al. (2013), Fig. 1C. Head-bob timing after training at 80 bpm.

Matching tempo differs from matching the beat

Tempo is the repetition rate of the sound sequence, expressed in beats per minute. Matching tempo means producing the same average number of movements per unit time. Two repeating events can share a tempo while occurring at different points within each cycle.

Phase describes the position of a movement relative to the sound cycle. Synchronization requires a stable timing relationship, ideally with the selected movement landmark near sound onset. The 2013 study measured both the movement rate and its alignment with individual beats.

At different unfamiliar tempos, Ronan maintained organized phase relationships even when her head reached its lowest point before or after the sound. These offsets distinguish successful coordination with the repeating stimulus from exact coincidence at the beat itself.



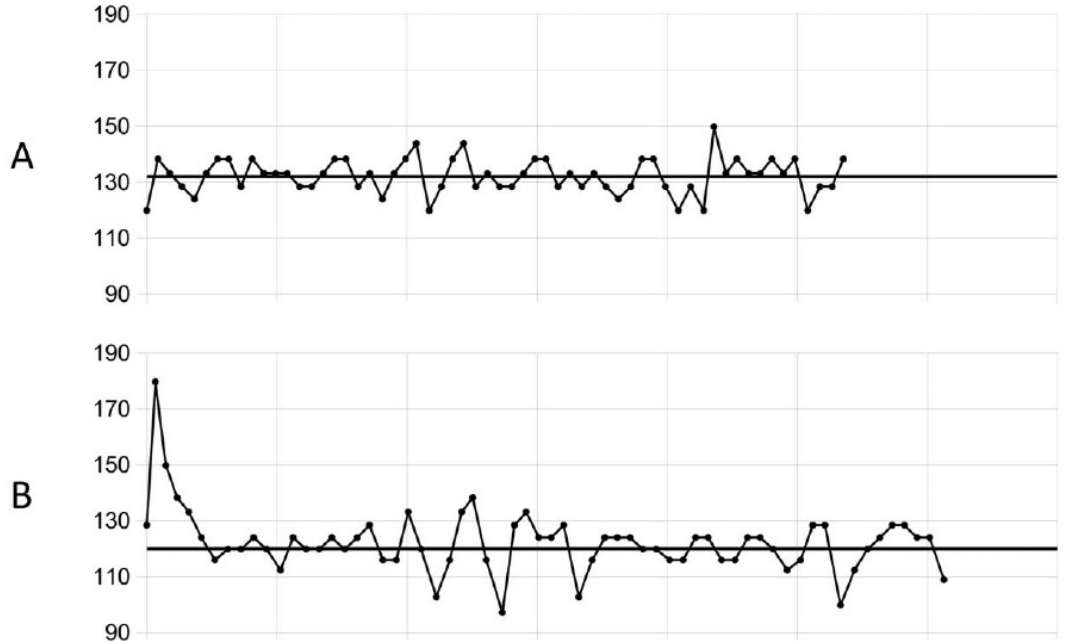
Cook et al. (2013), Fig. 2. Movement timing at five unfamiliar tempos.

The trained response is a continuous head bob

The behavioral response was an up-and-down movement of the head and neck, with roughly 10 cm or more of displacement during initial shaping. **Shaping** rewards successive approximations to a target action rather than waiting for the complete action to occur immediately.

Ronan first learned to bob in response to a visual cue. A brief whistle, previously associated with food, marked the rewarded behavior. The whistle was followed by a capelin, allowing the trainer to identify the successful action precisely before delivering food.

The resulting behavior was continuous across many sound events. Researchers could therefore compare successive movements with successive beats, rather than measuring a single reaction to a sound. Extended trials later tested whether this coordination persisted beyond the initial rewarded run.



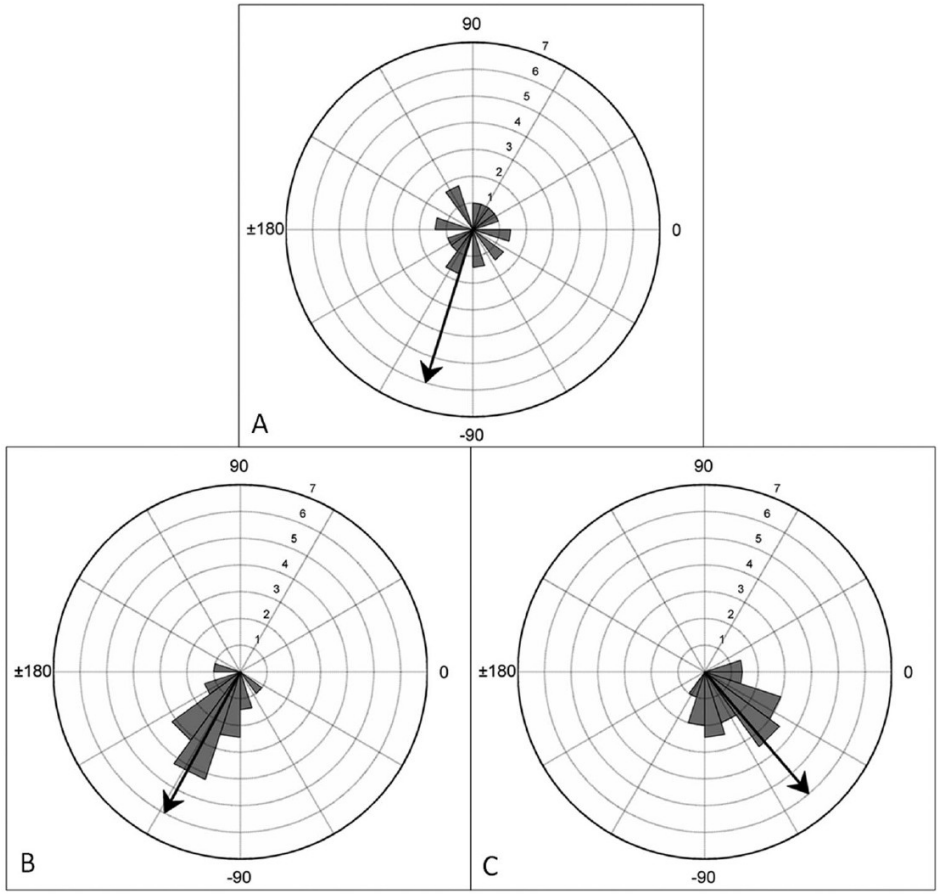
Cook et al. (2013), Fig. 6A–B. Successive movement rates during extended trials.

A blind separates sound cues from trainer movement

Ronan rested her foreflippers on a station while facing a wooden panel. This position left her head free to move. A reward assistant sat beyond the panel, and the experimenter watched from behind a separate blind to the animal's side.

Concealing the humans removed visible body movements that might supply an alternative rhythm. Food could be delivered through a small opening after the trial without exposing a moving trainer during the measured response. The acoustic sequence remained the intended timing cue.

Testing compared behavior before and after explicit entrainment training. Ronan initially failed to synchronize reliably with the presented rhythms. After training, her movements acquired organized relationships with the stimulus, linking the experimental history to the timing behavior observed on later trials.



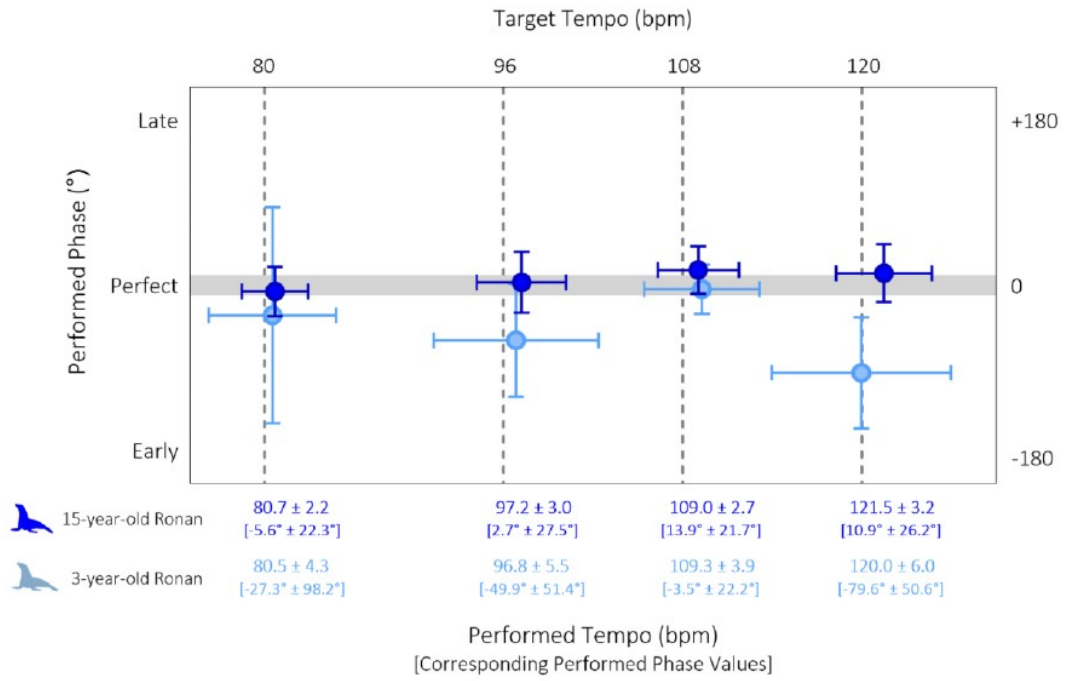
Cook et al. (2013), Fig. 1. Timing distributions for spontaneous and trained tests.

Rewarding speed alone produces variable timing

The initial training procedure rewarded Ronan when her bobbing rate approached the rate of either an 80 or a 120 bpm stimulus. During this stage, she produced highly variable movement rates and made little progress toward consistent synchronization with individual beats.

The trainers then separated the two tempos into different sessions and changed the criterion. Ronan was rewarded for bringing her head down on a beat, initially for a single movement. The required run was gradually increased to 20 consecutive synchronized movements.

This change trained an event-to-event relationship instead of only an average speed. The rapid improvement during the synchronized-bob stage provides a behavioral explanation for why a suitable reinforcement contingency mattered: the rewarded feature became timing relative to the sound.



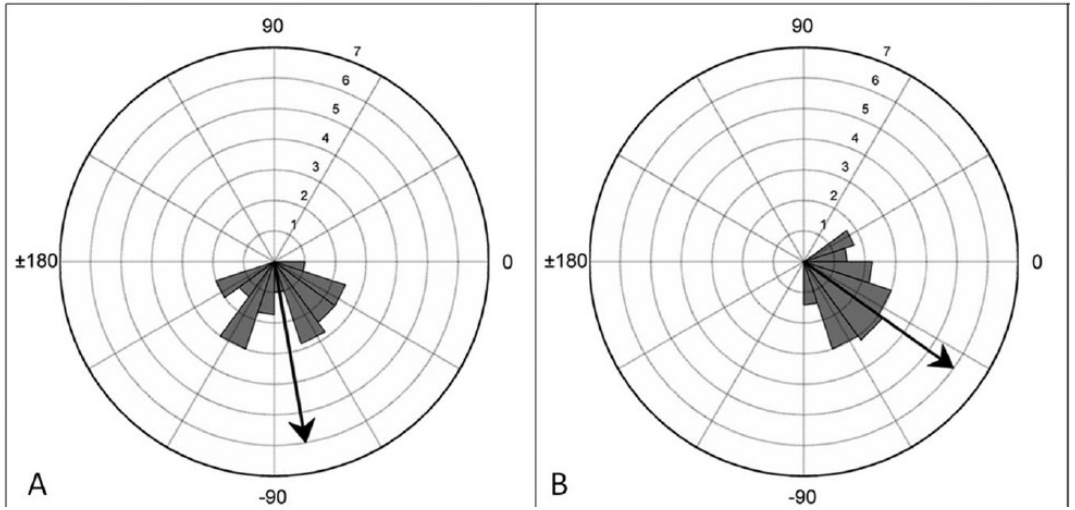
Cook et al. (2025), Fig. 2. Early and experienced performance across familiar rates.

Training two rates supports transfer to new rates

Ronan’s training used 80 and 120 bpm, but her transfer tests included 72, 88, 96, 108, and 132 bpm. **Transfer** means applying an acquired behavior to a stimulus condition that was not explicitly used to teach that behavior.

She coordinated head movements with all five unfamiliar rates on the first transfer presentations. Thus, successful performance was distributed across a tempo range rather than confined to the two trained examples. Both slower and faster rhythms could guide the repeated response.

The slower transfer conditions also exposed phase offsets: movement could remain consistently ahead of the sound rather than coinciding with it. Transfer therefore established flexible rate adjustment and stable coordination while retaining information about the particular timing relationship at each rate.



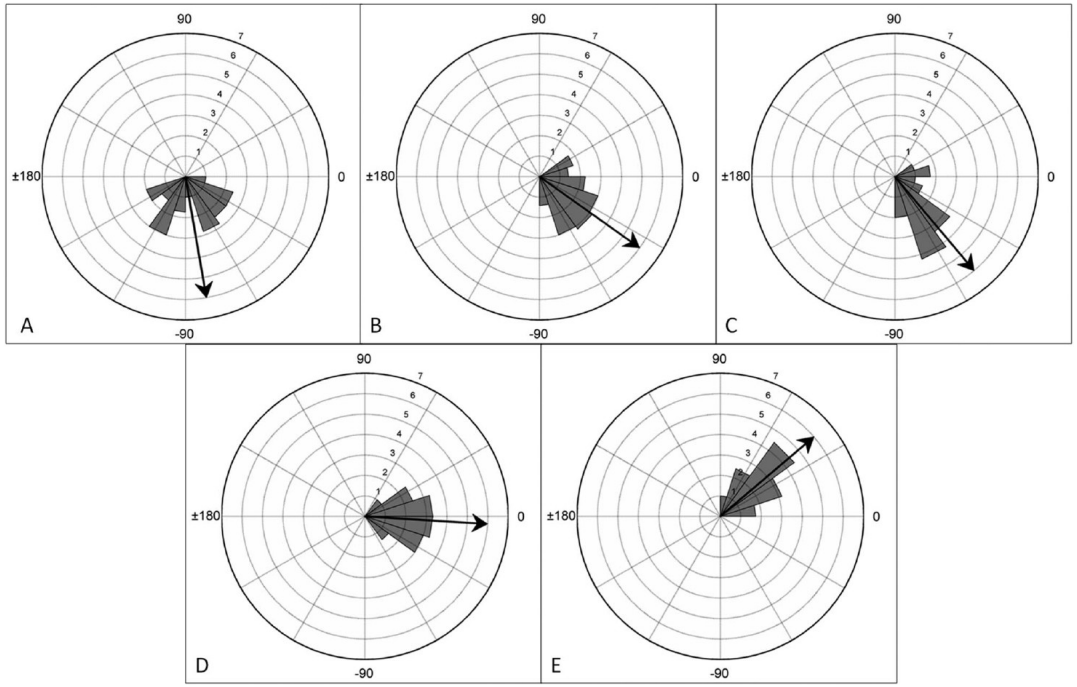
Cook et al. (2013), Fig. 2A–B. Transfer to 72 and 88 bpm.

Novel tempos change the movement–sound relationship

At 96 bpm, Ronan’s average movement rate was about 96.8 bpm, while at 108 bpm it was about 108.1 bpm. Her repeated actions followed the new sound rates closely despite the lack of explicit training at these particular tempos.

The position within the sound cycle also changed with tempo. At 108 bpm, the average phase was close to the beat, whereas faster and slower conditions showed larger systematic offsets. Matching the average rate and aligning precisely with onset are separable aspects of coordination.

A tempo-dependent phase relationship can arise when a repeating behavioral process is adjusted by the sound sequence. It motivates experiments that alter timing during a trial, because the transient response reveals how the animal corrects its ongoing movement after the external rhythm changes.



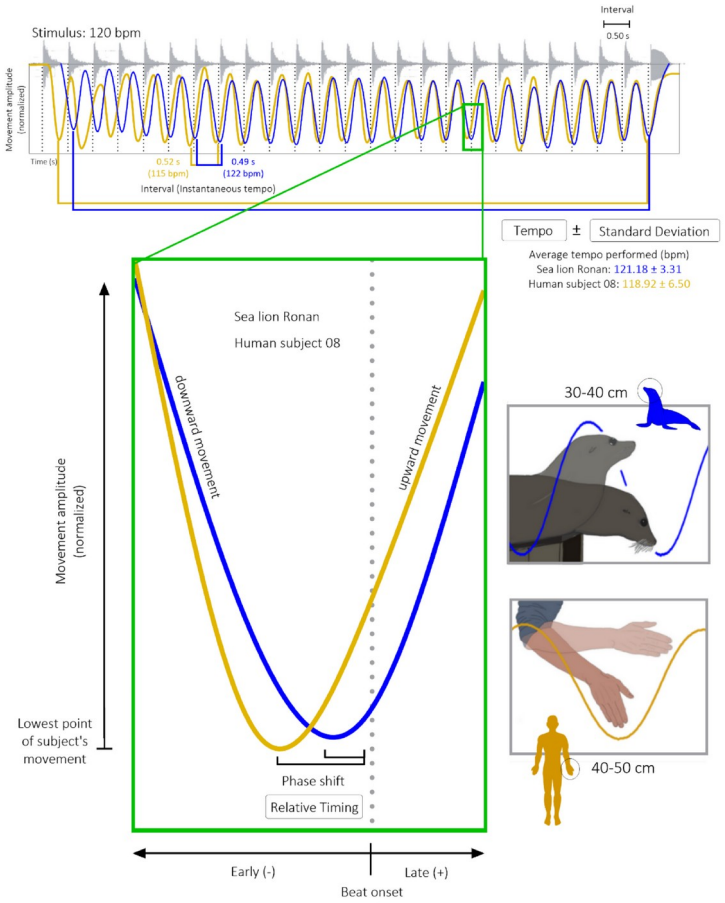
Cook et al. (2013), Fig. 2A–E. Transfer at 96, 108, and 132 bpm.

Movement landmarks make synchronization measurable

A movement unfolds continuously, but synchronization analysis needs a clearly specified event. In Ronan's task, the selected event was the lowest point of the head bob. In the later human comparison, it was the fullest downward extension of the arm.

Using a consistent landmark prevents arbitrary changes in what counts as a beat-marking response. The lowest point is compared with the sound onset, and the interval between successive low points supplies the movement's repetition rate. Rate and phase can then be evaluated separately.

The 2025 experiment recorded movements at 240 frames per second. Each frame represented roughly 4 ms, giving much finer timing resolution than ordinary video. Audio from the same recording supplied the sound onsets against which those movement landmarks were measured.



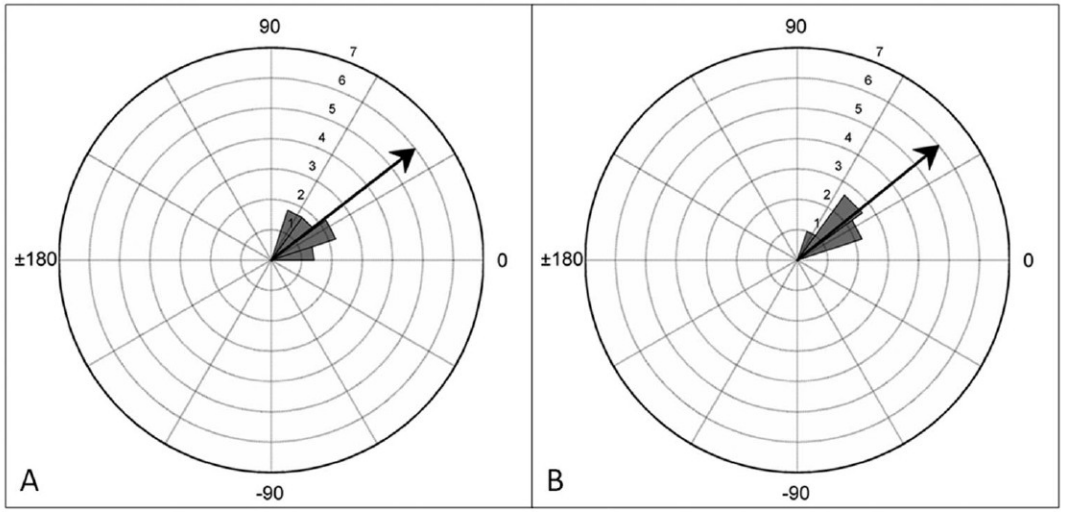
Cook et al. (2025), Fig. 1. Movement trajectories and beat-relative timing.

A missing sound tests continuation of the rhythm

An animal that simply reacts to each sound should change its behavior when a sound is omitted. Researchers therefore inserted occasional missing beats into a 120 bpm sequence, replacing the original continuous modulation with discrete clicks to leave genuinely silent gaps.

Ronan continued bobbing through the missing beats. The timing relationships of the movements immediately before and after a gap were similar, and she remained coordinated with the sequence rather than restarting the movement in response to the next available sound.

Continuation through an omission supports an ongoing timing process that extends beyond isolated sound-triggered reactions. A remembered cycle can carry the movement across a silent interval, while subsequent sounds provide new opportunities to adjust the relationship between the external rhythm and the behavior.



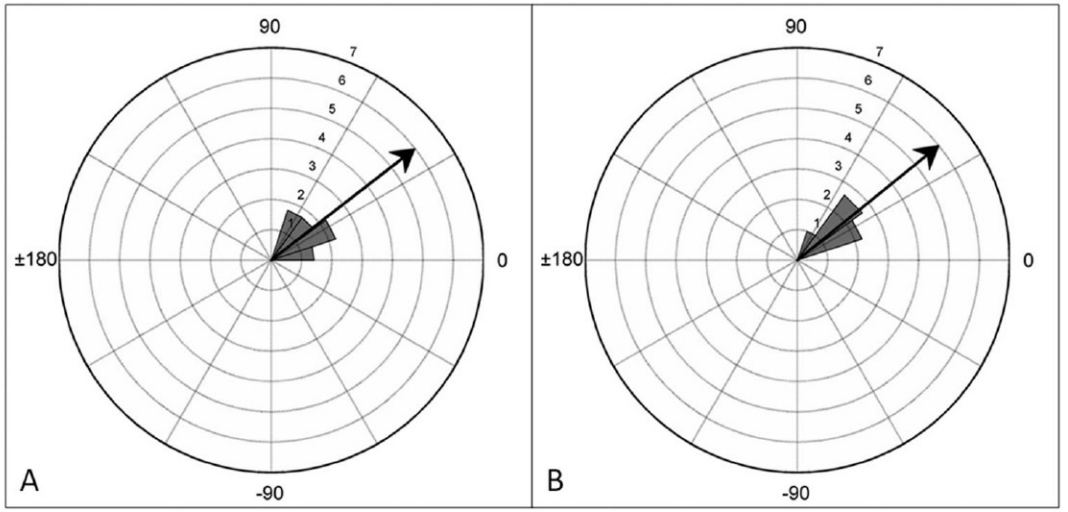
Cook et al. (2013), Fig. 3A–B. Movement timing around omitted beats.

Discrete clicks preserve coordinated movement

The omission test changed more than whether a beat occurred. Earlier tracks contained frequency modulation between beats; the omission stimulus used discrete clicks. Removing the continuous acoustic background made it possible to test behavior across an interval without a timing sound.

Ronan maintained entrained movement under this changed stimulus format. Her continued performance linked the head-bob rhythm to the regular temporal organization of the sequence rather than requiring the original continuous modulation as an indispensable acoustic cue.

This stimulus manipulation also separated the sound's detailed spectral pattern from its event timing. A response could follow a shared temporal regularity across different acoustic forms, supporting the use of later musical and percussive stimuli to test increasingly varied auditory–motor coordination.



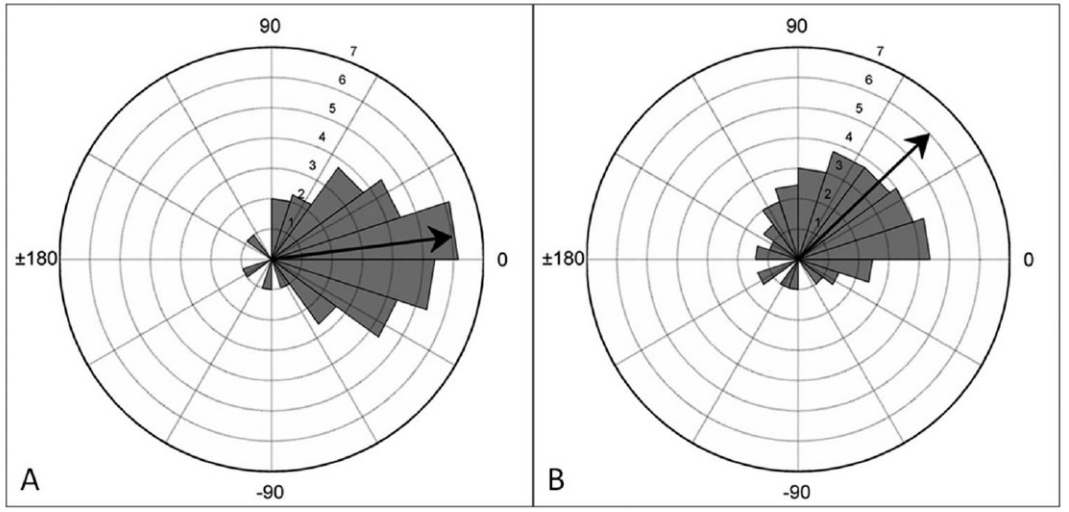
Cook et al. (2013), Fig. 3A–B. Timing before and after silent omissions.

Music introduces several competing sound events

A musical recording contains overlapping instruments, accents, and sounds between beats. To keep a steady beat, the animal must maintain a regular movement relationship amid these additional acoustic events rather than respond separately to every transient in the recording.

Ronan transferred to the song Everybody at about 108 bpm and to Boogie Wonderland at 130 bpm. Her head movements were organized around the musical pulse. The experiment therefore extended the response from simple artificial rhythms to temporally complex sound recordings.

A song can contain a stable underlying pulse even when its surface sounds are irregular. Successful transfer connected the trained repeated movement with that pulse, while differences between songs and trial histories revealed that experience with the particular musical material still influenced performance.



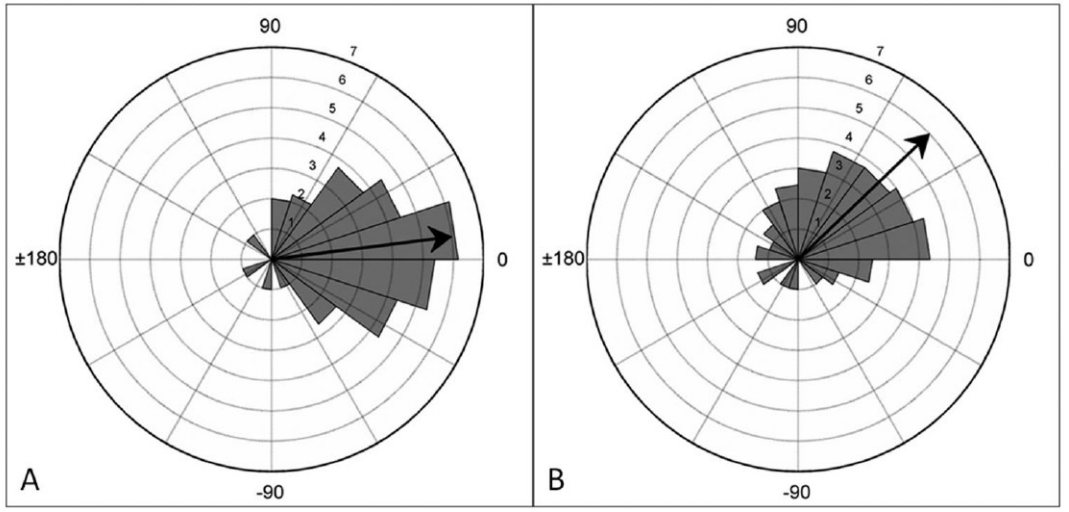
Cook et al. (2013), Fig. 4A–B. Timing during Everybody and Boogie Wonderland.

Prior exposure and reinforcement are different

Before its transfer test, Ronan had been exposed to Everybody without achieving criterion. She was subsequently trained with a short section of that song. Boogie Wonderland had also been presented briefly before transfer testing, but the initial exposure had not produced entrained bobbing.

These histories matter because an unfamiliar musical recording is not necessarily a completely unheard sound. The experiments distinguished earlier exposure, successful training, and subsequent test performance rather than treating all experiences with a song as equivalent learning episodes.

Ronan later reached the synchronized-bob criterion repeatedly with both songs. Maintaining the learned action with complex recordings required coordinating movement with a regular pulse despite the recording's additional events, and the training history supplied information about how that coordination became reliable.



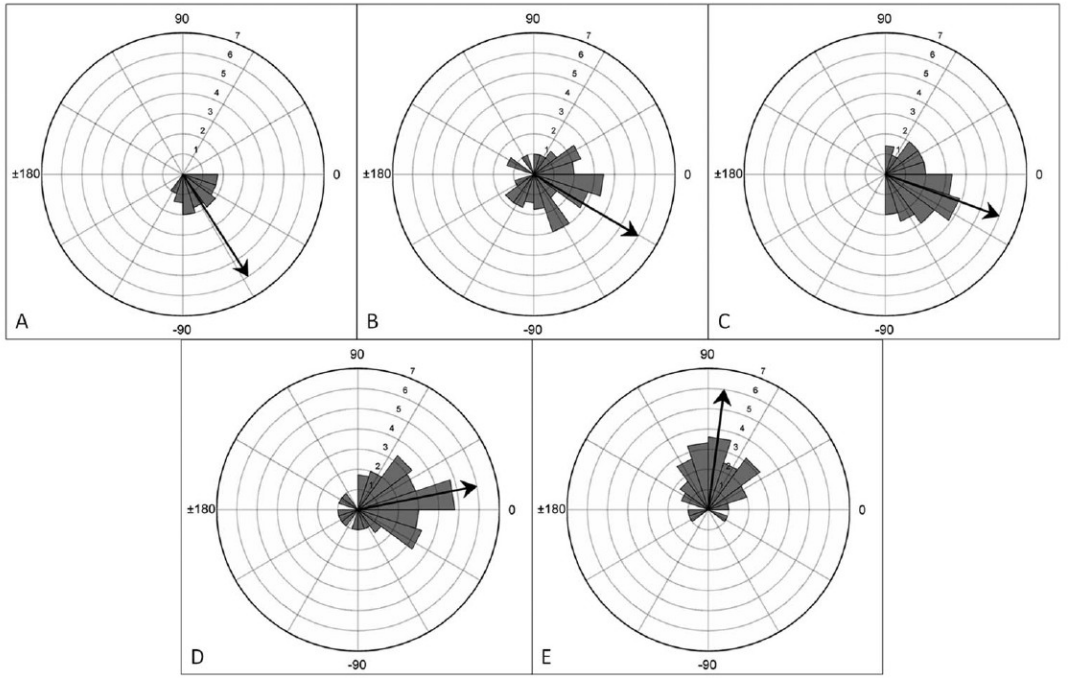
Cook et al. (2013), Fig. 4A–B. Successful timing in two music conditions.

Changing a song’s speed tests flexible pulse tracking

After training with Boogie Wonderland at its original tempo, researchers tested modified versions at 104, 117, 124, 137, and 143 bpm. The musical content remained related to the familiar song while the speed of the underlying pulse changed.

Ronan entrained at all five modified tempos. Successful trials were interspersed with trials using the original version, so transfer required adjusting to the current rate rather than continuing the rate rewarded on every preceding musical presentation.

This manipulation separated familiarity with a recording from the ability to track its temporal scale. The same song supported different movement speeds. At the fastest and slowest rates, the phase relationship could differ substantially from the native-tempo relationship while the movement remained organized.



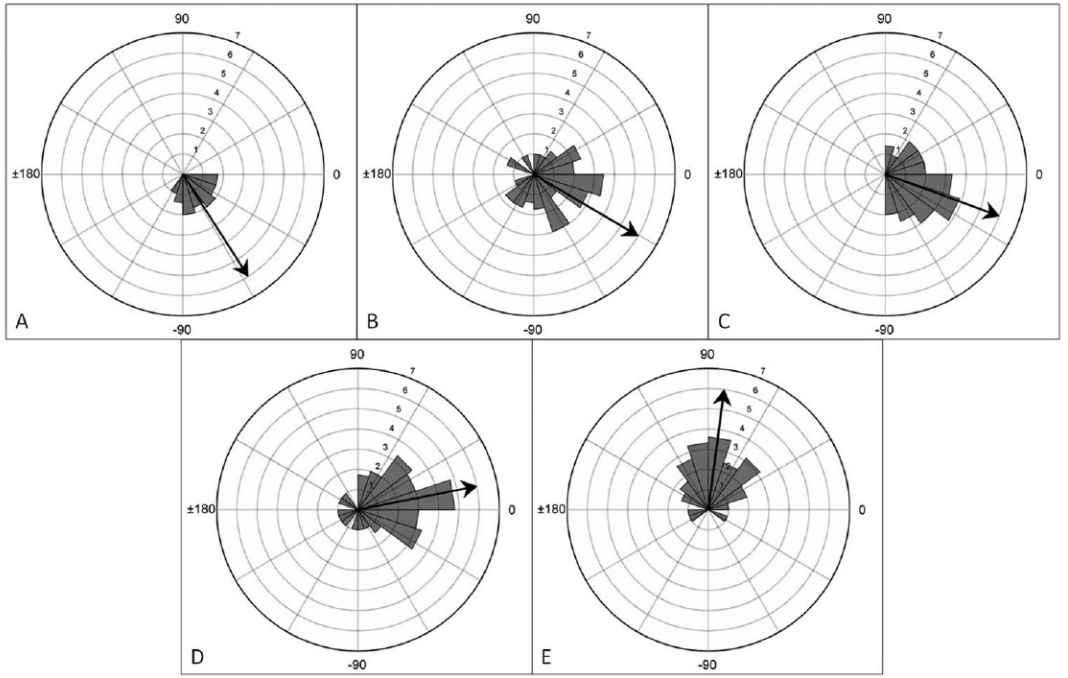
Cook et al. (2013), Fig. 5A–E. Transfer across altered Boogie Wonderland tempos.

Music transfer spans both slower and faster versions

The native Boogie Wonderland recording used in training was about 130 bpm. Transfer included slower versions down to 104 bpm and a faster version at 143 bpm, covering changes in both directions rather than only gradual increases in speed.

Ronan’s average head-bob rates moved with these changes. At the slowest tested musical rate, for example, her mean response was about 103.3 bpm. Repeated movements therefore followed the new pulse rather than remaining near the familiar 130 bpm motor rate.

Phase offsets also depended on the version. The organized distributions at each rate represented stable coordination, but their centers differed. These results provided the behavioral foundation for later perturbation tests that separated correction of the timing position from adjustment of the movement period.



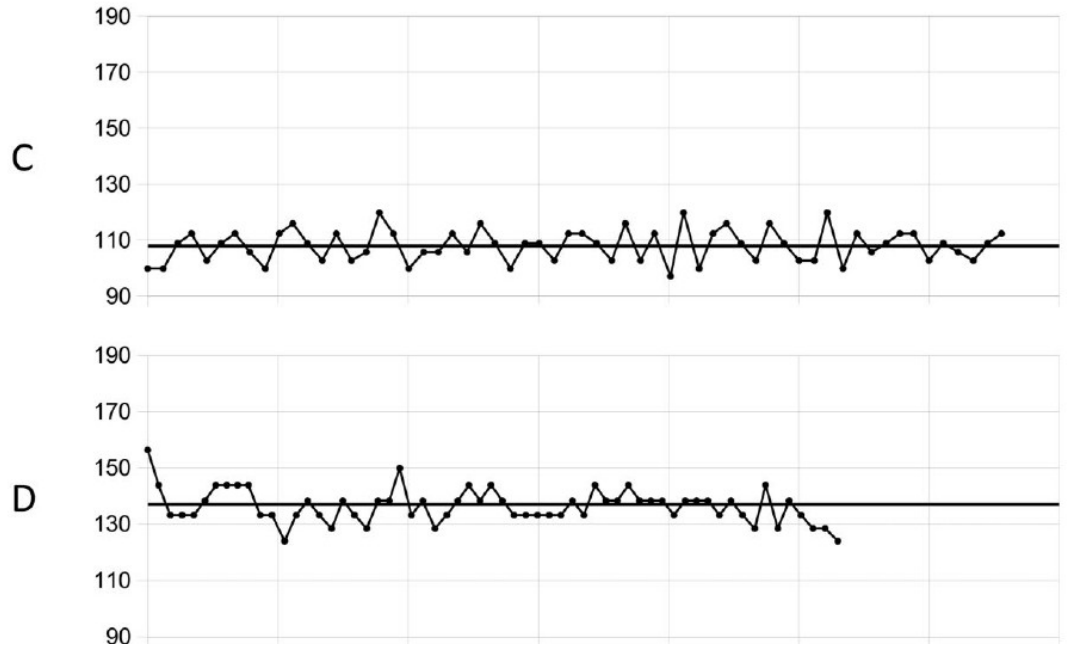
Cook et al. (2013), Fig. 5A–E. Movement phases across five musical tempos.

Long trials test persistence beyond the usual criterion

Most early trials ended after Ronan produced a rewarded run of 20 synchronized movements. Extended-duration tests increased the criterion to 60 beats, requiring her to maintain coordination for a substantially longer continuous interval before reinforcement was delivered.

The longer trials included simple stimuli and Boogie Wonderland at several rates. Ronan sustained organized timing across these sequences. Performance was therefore not confined to a short window immediately preceding a predictable reward at the usual trial length.

The trial trajectories also revealed adjustment near the beginning of a sound sequence. Individual head-bob rates could begin away from the stimulus rate and settle toward it, connecting the acquisition of coordination within a trial to its subsequent maintenance across many movements.



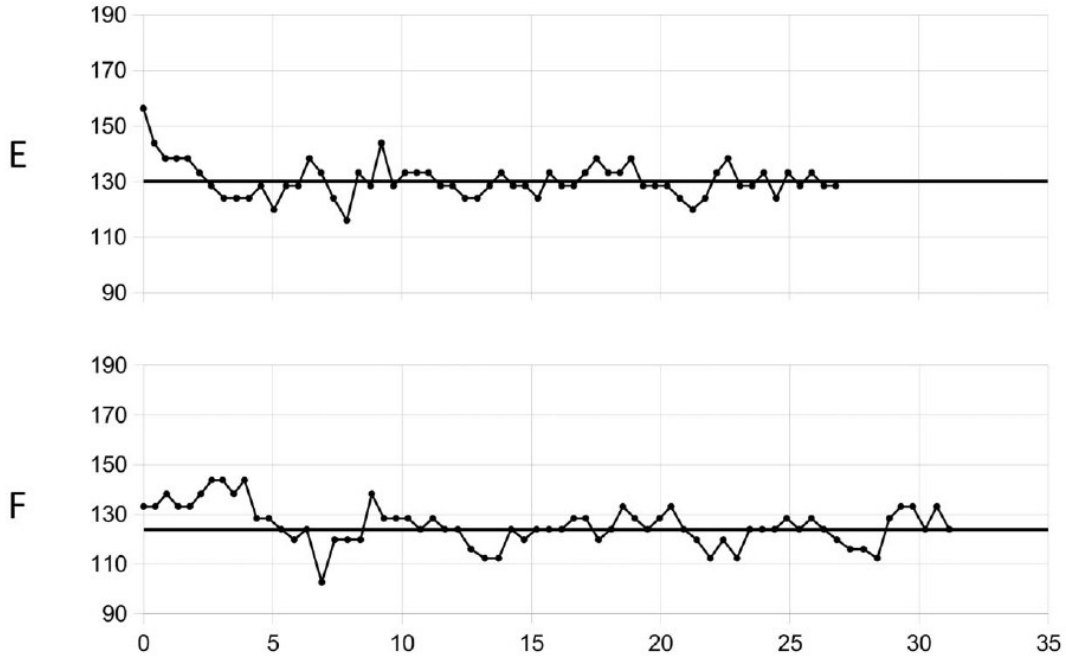
Cook et al. (2013), Fig. 6C–D. Movement-rate trajectories during extended musical trials.

A dancing reputation rests on controlled experiments

The public description of Ronan as a dancing sea lion refers to a trained head-bob response measured against sound timing. Her scientific contribution was not simply that she moved while music played, but that her movement adapted across rates and persisted through controlled tests.

Novel tempos, musical recordings, silent omissions, and extended trials each challenged a different simple explanation of the behavior. Together they connected the repeating motor response to temporal regularities that varied independently of the original two training examples.

Ronan’s study site was Long Marine Laboratory at UC Santa Cruz. The original researchers’ recordings are the relevant evidence for her timing performance; a publicity recording supplies behavioral context, while quantified trial measurements establish the accuracy, consistency, and transfer of that coordination.



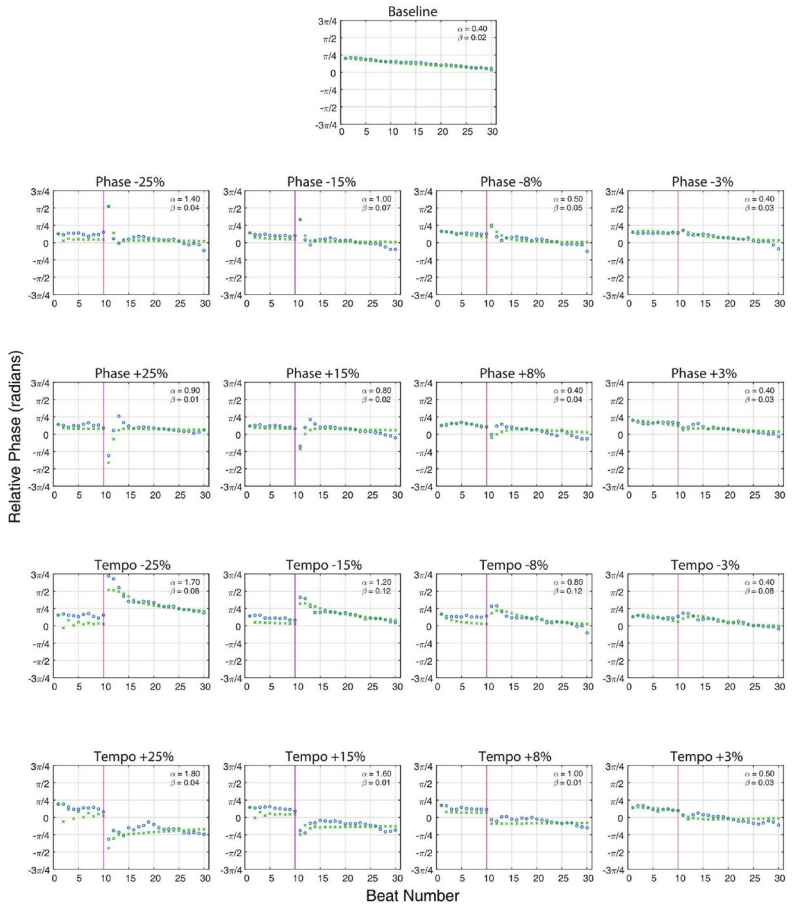
Cook et al. (2013), Fig. 6E–F. Extended coordination with music at 137 and 124 bpm.

Perturbations separate phase from period correction

A **phase perturbation** shifts where the sound sequence falls in time while preserving its eventual repetition rate. A **tempo perturbation** changes the interval between successive sounds. These manipulations challenge different components of an already coordinated movement.

Rouse and colleagues introduced one unexpected change during a trial, after a stable initial run of beats. Ronan then had to modify the movement she was already producing. The response immediately after the change exposed the dynamics of correction.

A repeating motor process can correct timing position quickly while changing its preferred period more slowly. Comparing phase and tempo perturbations therefore tested the proposed coupled-oscillator explanation with transient behavior, extending the evidence beyond stable performance measured after coordination had already been achieved.



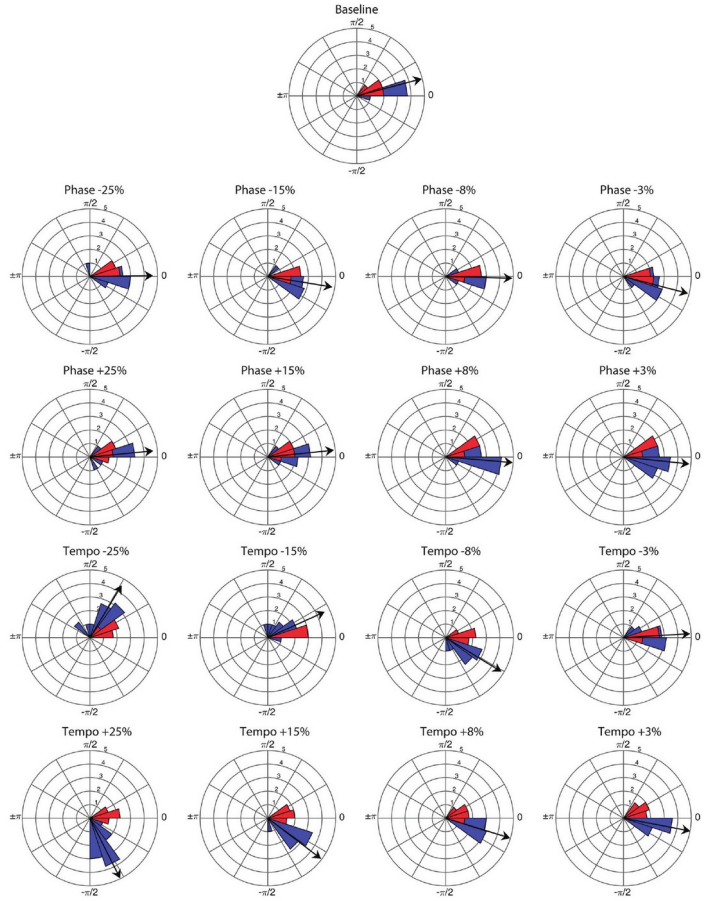
Rouse et al. (2016), Fig. 3. Responses to mid-trial timing changes.

The perturbation stimulus uses brief paired tones

The stimulus clicks combined tones at 659 and 1319 Hz for 10 ms. Repeating these brief events created a clear temporal sequence. The main baseline rate was 85 bpm, a rate Ronan had not previously encountered in training or testing.

Perturbations changed a timing interval by 3, 8, 15, or 25 percent in either direction. The change occurred between the sixteenth and twenty-fifth beat, with its position varied across conditions so that a fixed point in the sequence would not predict it.

At the baseline rate, the normal interval was about 706 ms. Controlled changes to that interval introduced experimentally specified timing errors into a sequence that Ronan was already following, allowing the subsequent head bobs to reveal how she recovered coordination.



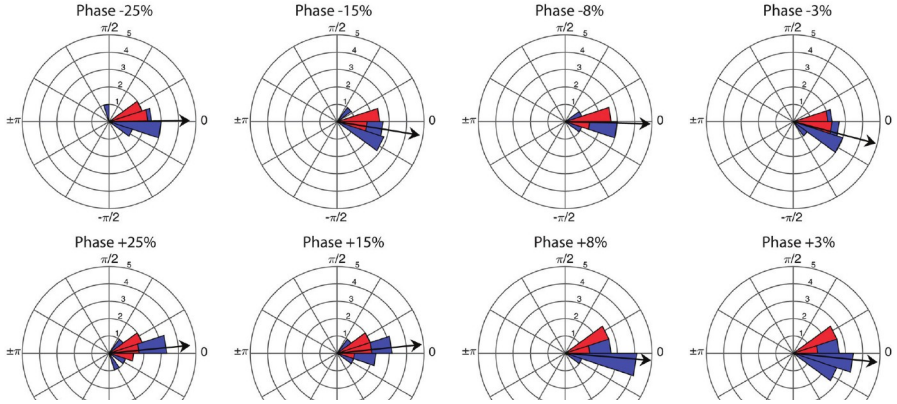
Rouse et al. (2016), Fig. 1. Phase distributions before and after timing changes.

A phase shift moves the beat without a new tempo

In a phase-shift trial, a sound occurs earlier or later than expected, but later sounds return to the original spacing. The stimulus's temporal position changes while its continuing tempo remains fixed, separating an onset-position error from a lasting change in speed.

Ronan adjusted her head-bob timing after both advances and delays. Her movement regained an organized relationship with the shifted sequence within a few beats, rather than staying locked to the timing established before the perturbation.

This recovery is **phase correction**, the adjustment of an ongoing cycle's position relative to an external event. Because the subsequent sound interval returns to baseline, successful recovery can be achieved by a transient movement adjustment without adopting a permanently different repetition rate.



Rouse et al. (2016), Fig. 1, phase conditions. Timing distributions after phase advances and delays.



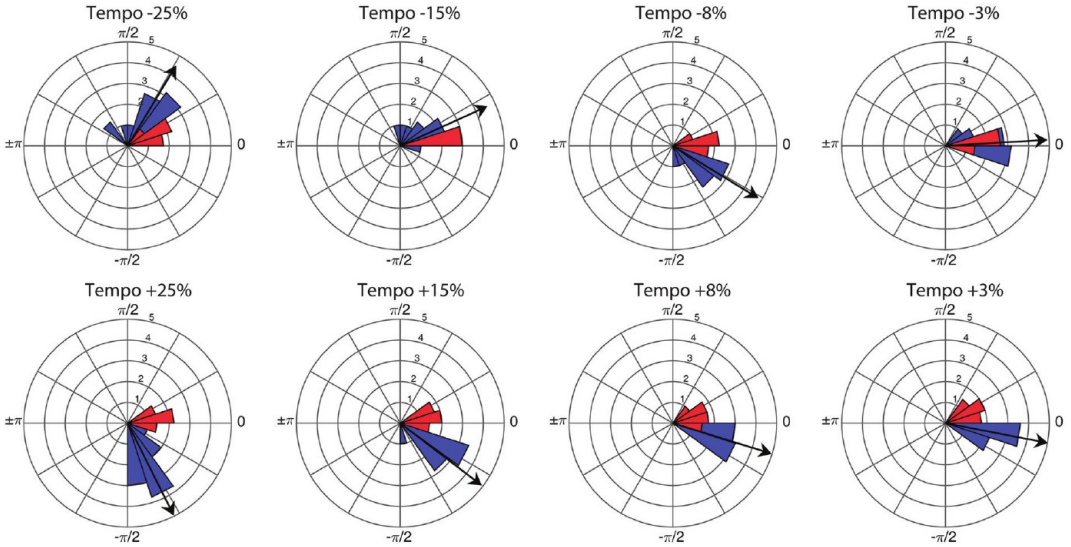
Rouse et al. (2016), Supplementary Video 1. Ronan adjusts to phase and tempo changes.

A tempo shift changes every later sound interval

A tempo perturbation alters the spacing of every subsequent beat. If the head-bob rhythm kept its original period, the relative position of movement and sound would drift increasingly across the trial, even after the first timing error had been corrected.

Ronan adapted after all the tested tempo changes. The response had to address the continuing mismatch in cycle duration, not merely the placement of a single unexpectedly early or late sound within an otherwise unchanged sequence.

The authors' sign convention referred to interval length: a positive interval change made the beat slower, while a negative change made it faster. Keeping interval and tempo distinct is necessary for interpreting which perturbations required Ronan to slow her repeated movement and which required her to accelerate it.



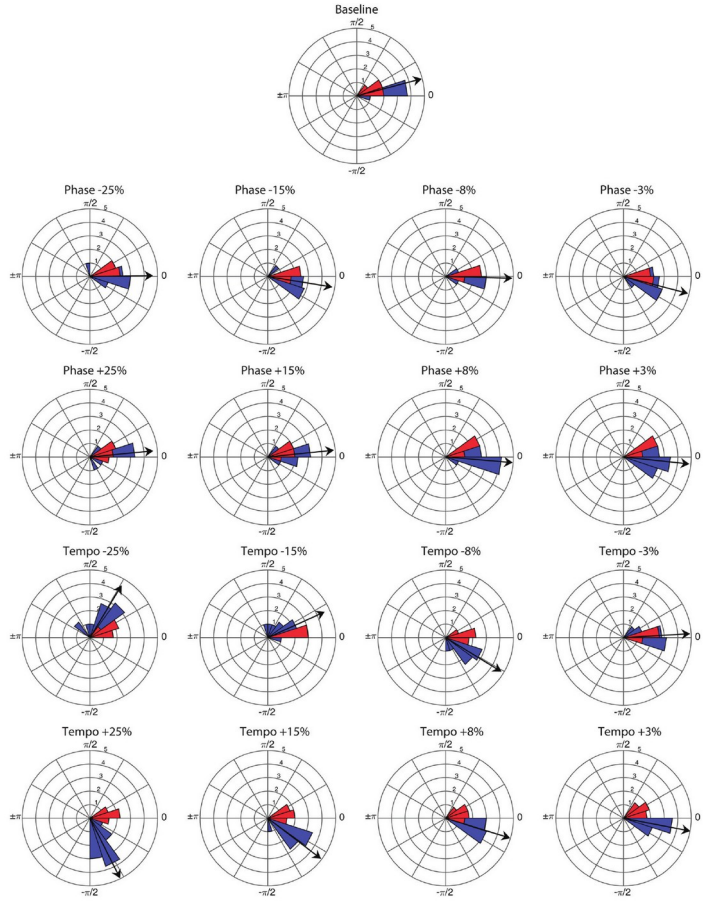
Rouse et al. (2016), Fig. 1, tempo conditions. Timing distributions after persistent rate changes.

The first transfer response already tracks the change

The first presentation of each perturbation provided a test before repeated experience with that exact change. Ronan retained organized movement relative to the beat after these first encounters, including phase changes and persistent changes in the stimulus interval.

The response therefore transferred beyond a repeatedly practiced correction at one fixed perturbation. The animal applied its existing auditory–motor coordination to an unfamiliar mid-trial disturbance, with later sounds continuing to influence the repeating behavioral cycle.

Initial-transfer data and repeated-trial data served different purposes. The former established that correction could appear at first exposure; the latter characterized the consistency and temporal course of correction across presentations, supporting a more detailed model of the behavior’s adjustment dynamics.



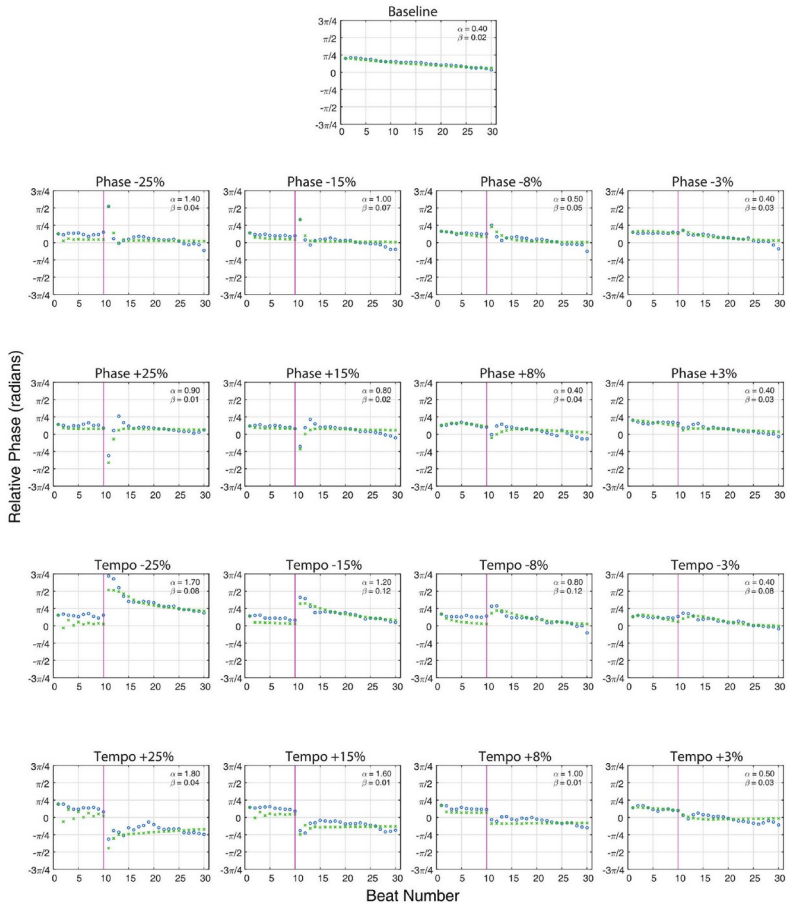
Rouse et al. (2016), Fig. 1. First presentations of timing-perturbation conditions.

Phase recovery occurs over successive movements

The researchers aligned trials at the perturbation and compared the relative phase of successive head bobs. Before the change, movement and sound were already coordinated. The sudden phase disturbance created a measurable error that could then be followed over later cycles.

Ronan corrected the disturbance within a few beats. Recovery was a trajectory across successive movements rather than an instantaneous switch from one static state to another. This time course supplied information about the strength and direction of adjustment.

The published model predicted these transient trajectories using coupling between the sound sequence and the behavioral oscillator. **Coupling** means that the timing of one process changes the state of the other; here, the sound events adjusted the timing of the continuing head-bob cycle.



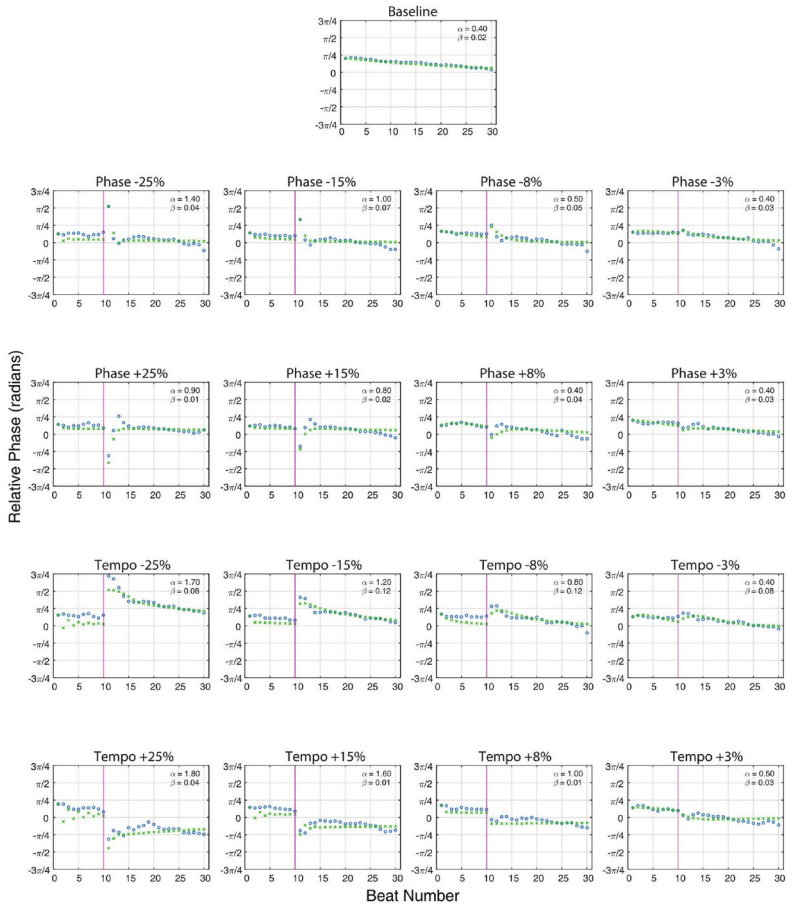
Rouse et al. (2016), Fig. 3. Beat-by-beat recovery after phase changes.

Rate changes produce a lasting timing adjustment

Following a lasting rate change, Ronan's initial movement period and the new stimulus period differed. Head movements therefore had to adjust across several cycles to approach a stable relationship with the new continuing sound sequence.

The measured trajectories recovered after changes in either direction. A faster stimulus required more rapid movement, whereas a slower stimulus required longer intervals between selected movement landmarks. The correction preserved a regular relationship between sound and action despite the disturbance.

The period component of the coupled-oscillator model represented adjustment to this continuing rate mismatch. Combining period adjustment with phase adjustment improved the fit to the measured recovery trajectories, connecting two distinct timing operations to the transient response produced by the animal.



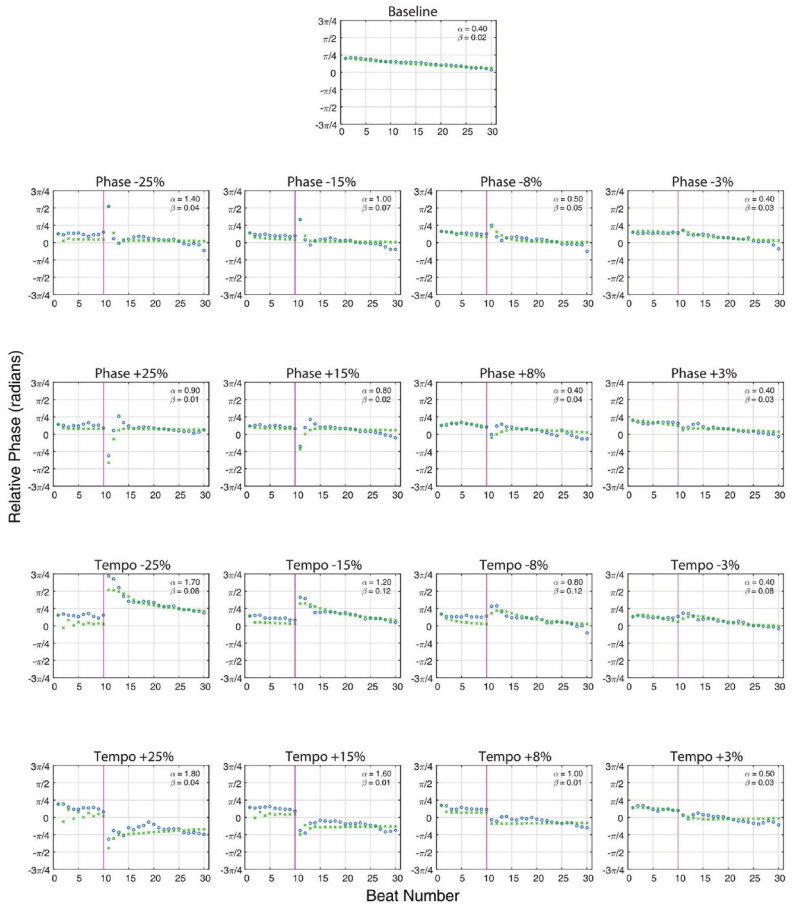
Rouse et al. (2016), Fig. 3. Recovery trajectories following rate shifts.

A double bob exposes an immediate motor adjustment

During some large slowing perturbations, the next sound occurred when Ronan's head was near the high point of a movement rather than its low point. She sometimes made an additional small nod before slowing the broader movement cycle.

The authors called these responses **double bobs** because two identifiable movements occurred around a beat. Their observation connected the timing disturbance with a specific change in movement form, rather than treating all behavior as identical cycles with only altered timestamps.

These unusual nods were excluded from the primary cycle-timing analysis and documented in Supplementary Video 1. Their occurrence is still informative behavioral material: an unexpected sound can modify both the timing of a repeated action and the shape of the immediate motor response.



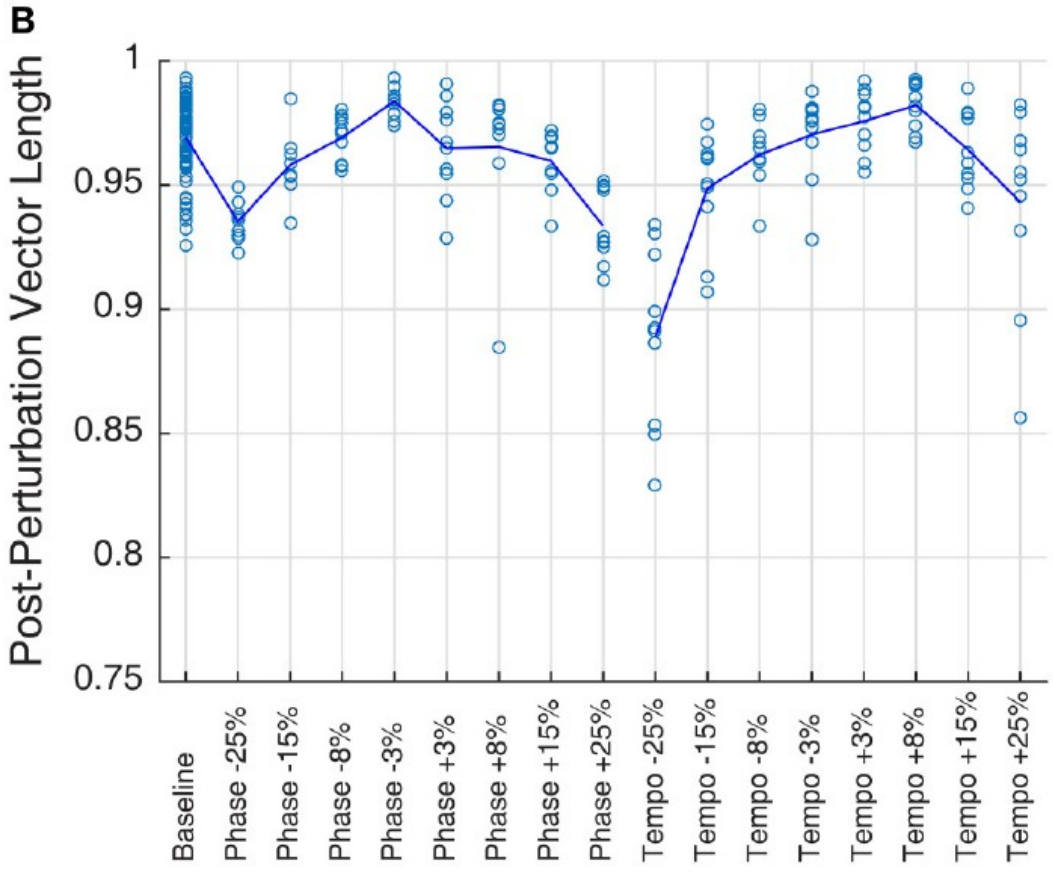
Rouse et al. (2016), Fig. 3. Behavioral timing after persistent interval changes.

Timing concentration and timing offset are separate

A series of movements can be consistently early, consistently late, or widely scattered around the beat. The average phase describes the center of that relationship, whereas the concentration of phases describes how consistently successive movements occupy that position.

Ronan's post-perturbation phases were tightly concentrated across the tested conditions. Some conditions nevertheless retained systematic offsets. High consistency therefore describes reliable coordination with the sequence and does not automatically mean that the chosen movement landmark coincides exactly with sound onset.

The researchers evaluated these properties separately because different aspects of the timing process influence them. Correcting a disturbance may stabilize a relationship before eliminating its offset, and a repeating behavioral process can remain strongly coordinated while occupying a stable position ahead of or behind the sound.



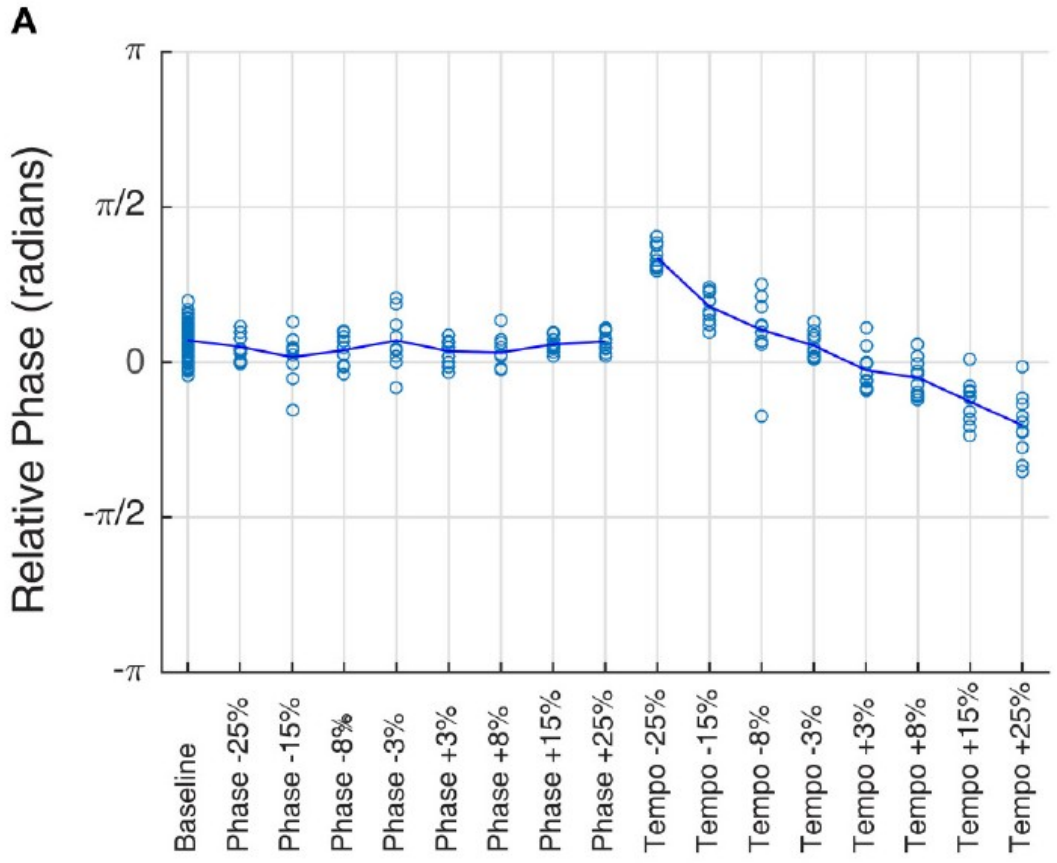
Rouse et al. (2016), Fig. 2B. Consistency of beat-relative movement timing.

The stable phase varies with the new tempo

Phase changes and tempo changes produced different patterns after recovery. Under phase perturbations, the long-term stimulus rate returned to its original value. Under tempo perturbations, the ongoing sound interval remained altered, creating a new continuing rate relationship.

Ronan's average post-perturbation phase varied systematically with the altered tempo. Slower and faster conditions therefore retained distinct timing positions even when the head movements had regained a consistent relationship with the continuing sequence.

This phase–tempo relationship matched a pattern seen in the earlier transfer experiments. The repeated finding linked stable coordination across unfamiliar tempos with the transient correction of a sudden rate change, giving a common behavioral constraint for the oscillator model to explain.



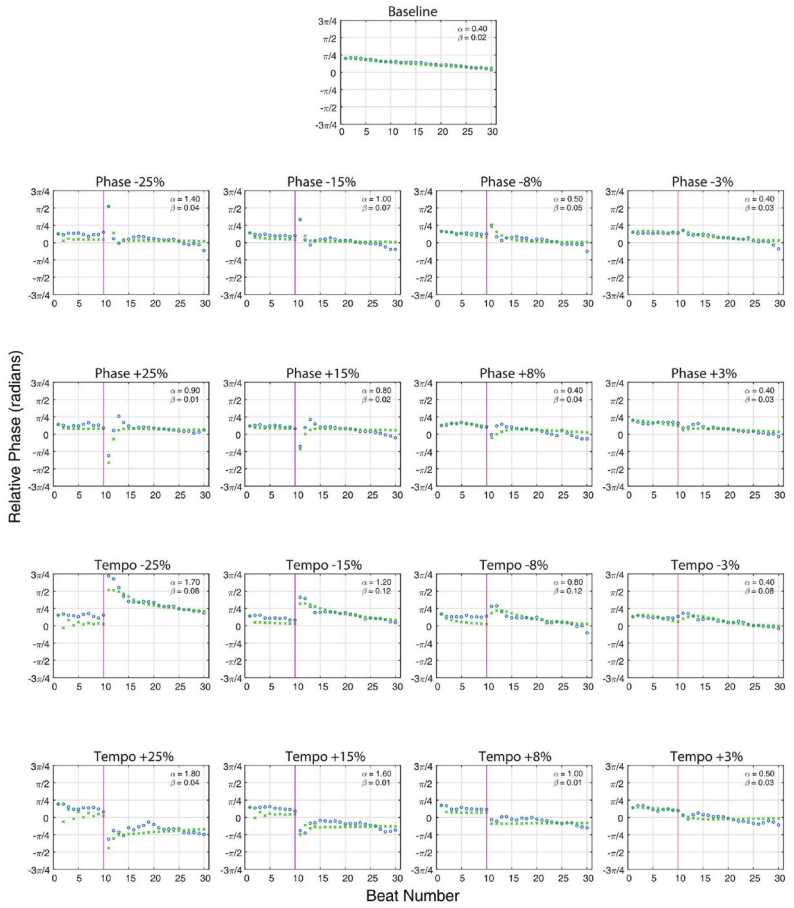
Rouse et al. (2016), Fig. 2A. Average offsets after phase and tempo perturbations.

An oscillator is a repeating timing process

An **oscillator** is a process that repeatedly passes through a cycle. The model represented Ronan's head-bob timing as an oscillator influenced by the recurring auditory events. Its cycle position and repetition period could both change as sounds arrived.

The hypothesis was that coupled timing processes could account for the observed coordination. A sound arriving at an unexpected cycle position generated a corrective influence, while continuing differences between movement and sound periods produced further adjustment across subsequent cycles.

The published model closely matched the measured recovery trajectories for phase and tempo disturbances. This agreement supplied a mechanistic description of timing behavior, while the experiment itself measured head movement and sound timing rather than recording the neurons that might implement those coupled processes.



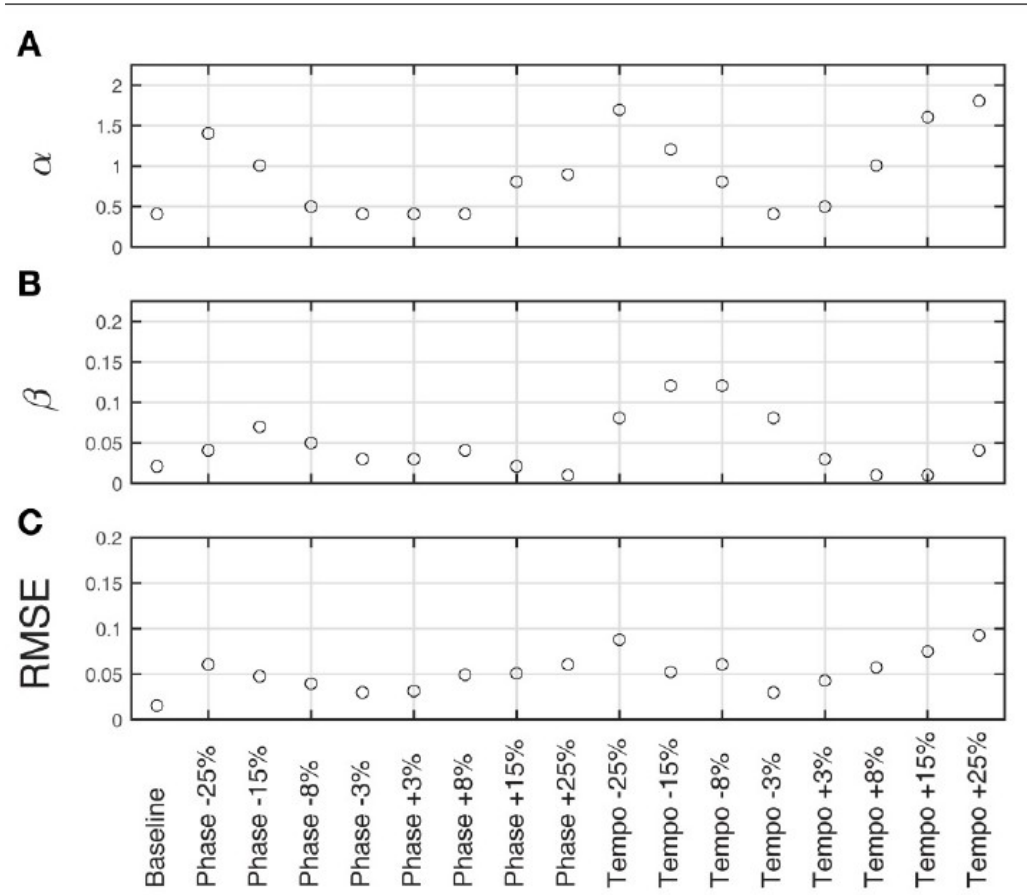
Rouse et al. (2016), Fig. 3. Measured recovery and coupled-oscillator predictions.

Strong phase coupling restores the timing position

The model's **phase-coupling parameter** described how strongly a sound event adjusted the position of the behavioral cycle. Stronger coupling produced a larger correction for a given timing error, allowing the movement to regain a stable relationship over fewer cycles.

Across the tested perturbations, phase adjustment was much stronger than period adjustment in the fitted model. Ronan's behavior could therefore be described primarily by rapid correction of cycle position, supplemented by weaker changes in the underlying repetition period.

The strength of phase coupling also increased with the magnitude of the disturbance. Larger timing errors elicited stronger corrections, connecting the severity of the unexpected auditory change with the modeled adjustment rather than assigning a single identical correction to every perturbation.



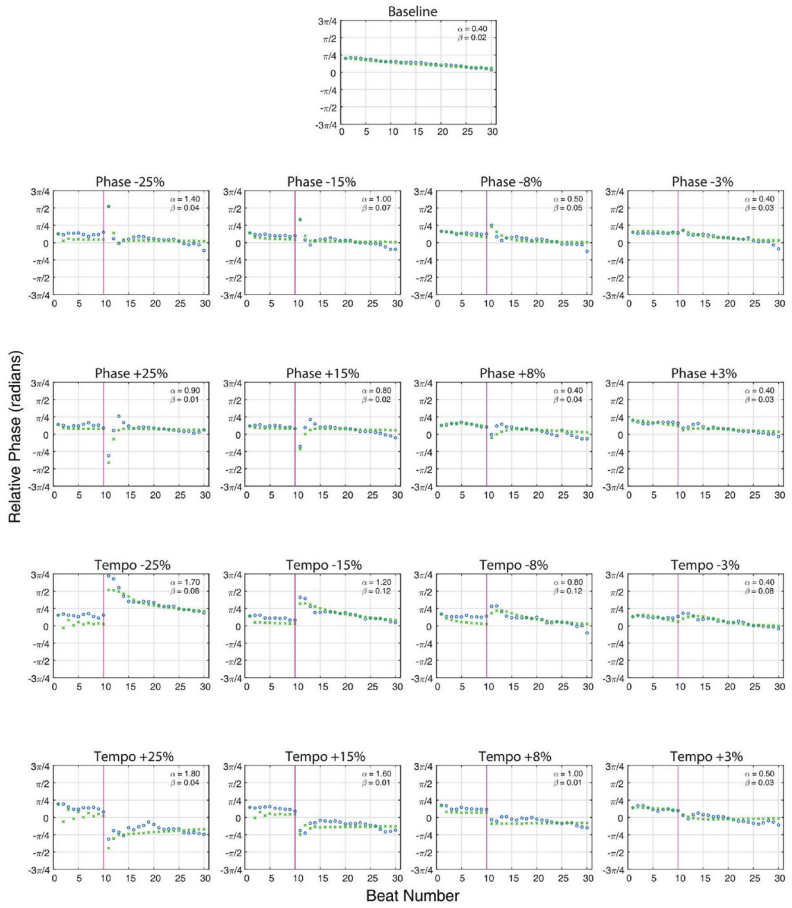
Rouse et al. (2016), Fig. 4A–C. Fitted phase and period adjustment parameters.

Period coupling adjusts the ongoing cycle duration

The **period** is the duration of one complete movement cycle. Changing the period alters the repetition rate itself. This differs from briefly shifting the cycle's position while leaving the preferred duration of later cycles unchanged.

In the fitted model, period coupling was weaker than phase coupling but improved the account of the observed behavior. Lasting rate changes required ongoing adjustment because correcting the first displaced event alone would leave later movements drifting relative to the sound.

The combination offered a behavioral mechanism with two time scales: relatively strong immediate phase correction and weaker period adaptation. These were model parameters inferred from movement trajectories, not measured concentrations of a transmitter or direct observations of separate neural populations in the sea lion.



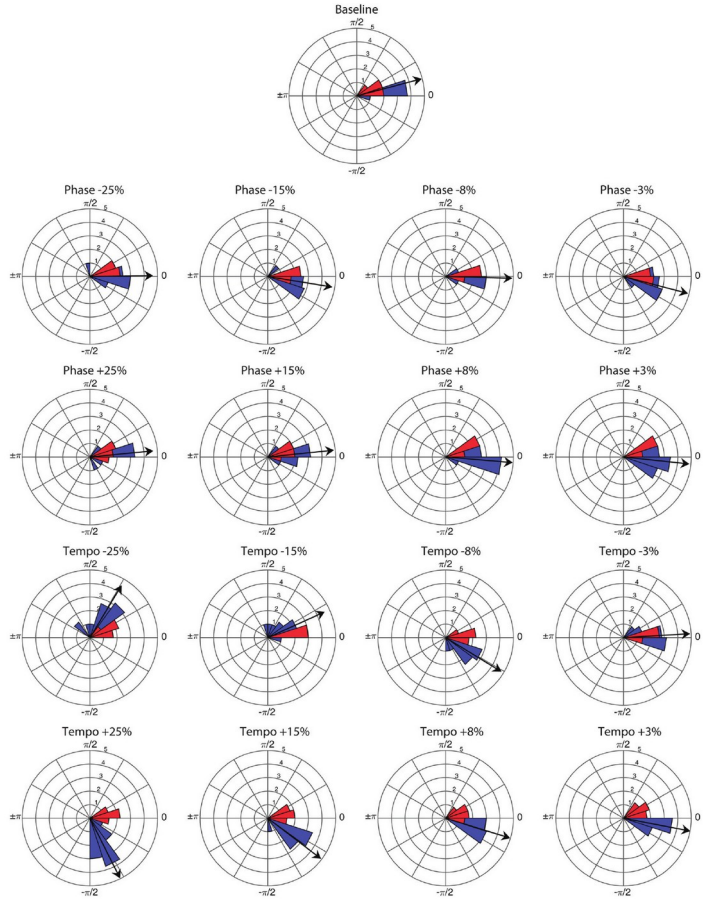
Rouse et al. (2016), Fig. 3. Modeled and measured adaptation to new rates.

The same correction transfers to other base rates

The main perturbation series used 85 bpm. Researchers also tested two additional unfamiliar baseline rates, approximately 77.3 and 94.4 bpm, with phase and tempo disturbances. This extended the correction test beyond a single preferred starting rhythm.

Ronan adapted across the additional baseline conditions as well. The ability to correct an ongoing response therefore generalized both across disturbance types and across the initial rate from which the disturbance was introduced.

A timing mechanism that operates across these conditions must use the relationship between the current sound interval and the current movement cycle. A correction learned only for a particular sound timestamp or one fixed movement rate would not provide the same flexibility across different baseline rhythms.



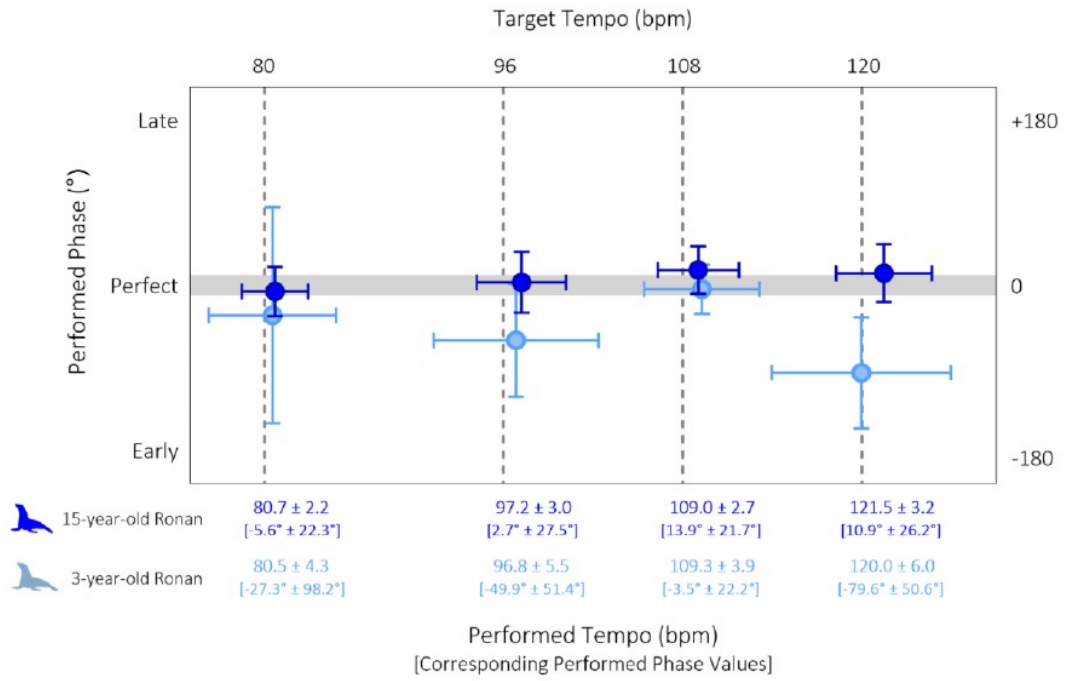
Rouse et al. (2016), Fig. 1. Controlled timing perturbations around a baseline rhythm.

Experience changes the animal’s synchronization profile

The 2025 study revisited Ronan after years of intermittent beat-keeping experience. Her movements were compared at 80, 96, 108, and 120 bpm, using familiar simple stimuli related to those in the original experiments.

By age 15, her movement rates and phases were generally closer to the stimulus than at age 3, with more consistent timing across conditions. The long-term comparison therefore changed the behavioral description of an individual previously characterized by more substantial tempo-dependent offsets.

Experience and development both occurred during the intervening years. The measured change was improved performance in an experienced adult, rather than an experiment that independently manipulated age and practice. That longitudinal history is necessary when comparing her current capacity with the earlier benchmark.



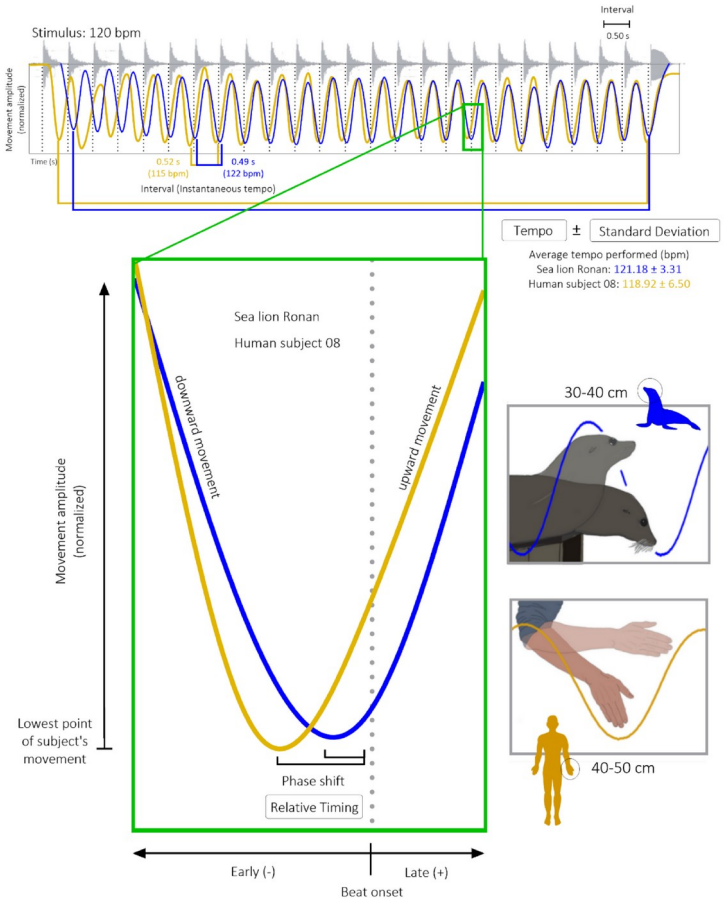
Cook et al. (2025), Fig. 2. Performance at ages 3 and 15 years.

The human task matches a large-amplitude movement

Earlier comparisons often contrasted sea-lion head bobs with human button presses. The 2025 experiment instead asked human participants to make a comfortable downward arm motion, bringing the selected movement landmark into alignment with the sound.

Humans held an arm beyond a panel and kept their gaze on a fixed mark, while the experimenter remained outside the room during trials. Ronan continued to work at her station with humans concealed behind a panel and blind.

The tasks shared sound sequences, large continuous movements, and a defined downward landmark. Matching these properties made the behavioral comparison more informative than contrasting a sea lion's whole-head movement with a small discrete finger action measured under unrelated stimulus conditions.



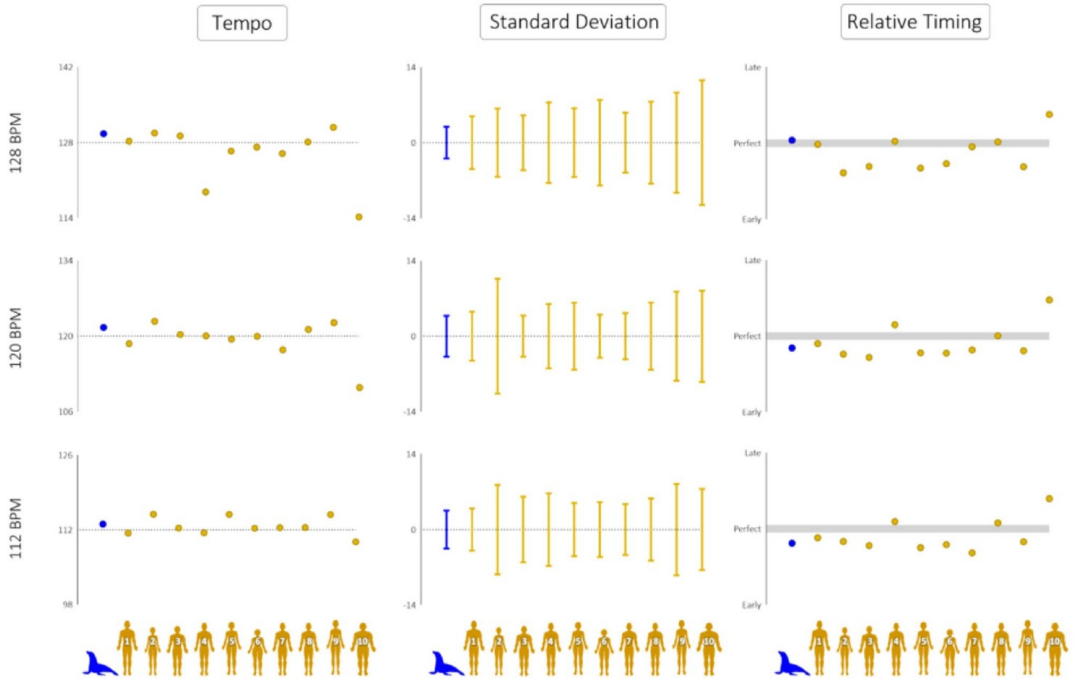
Cook et al. (2025), Fig. 1. Comparable downward movement landmarks.

A novel sound tests coordination beyond tonal tracks

The comparison stimulus was a repeated snare-drum sound, different from Ronan's familiar tonal tracks. Before testing, she received brief exposure at 96 bpm. This introduced the sound itself before the unfamiliar-rate comparisons began.

Both sea lion and humans then completed the same warm-up presentations and test sequences. The test rates were 112, 120, and 128 bpm. The middle rate was familiar to Ronan, while the two surrounding rates were novel.

Changing the sound's acoustic character while preserving regular event timing challenged the response to generalize across timbre as well as tempo. **Timbre** is the sound quality associated with its spectral and temporal structure, distinguishing a snare hit from the earlier paired-tone stimuli.



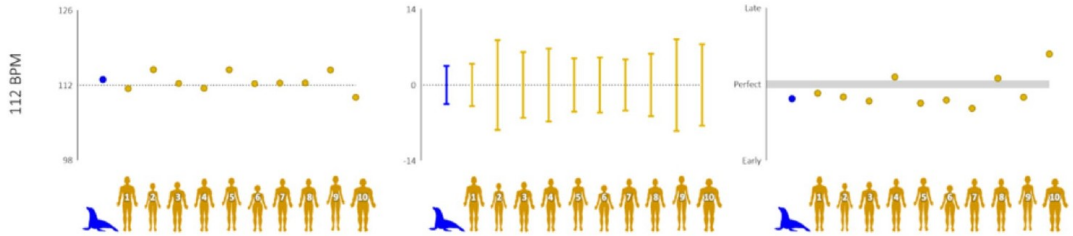
Cook et al. (2025), Fig. 3. Sea-lion and human performance with snare beats.

Novel rates flank a familiar training tempo

At 112 bpm, the snare stimulus was slower than Ronan’s familiar 120 bpm training rate. At 128 bpm, it was faster. Testing on either side of the practiced rate examined whether synchronization remained flexible rather than being confined to that particular learned tempo.

Ronan produced one head bob for each beat on the tested sequences. Her average performed rates were approximately 113.1, 121.6, and 129.0 bpm for the 112, 120, and 128 bpm stimuli, respectively.

These values show close rate matching across the novel acoustic and temporal conditions. The phase measurements supplied an additional dimension: the animal could maintain a regular repetition rate while reaching the selected movement landmark slightly before or after the sound, depending on the condition.



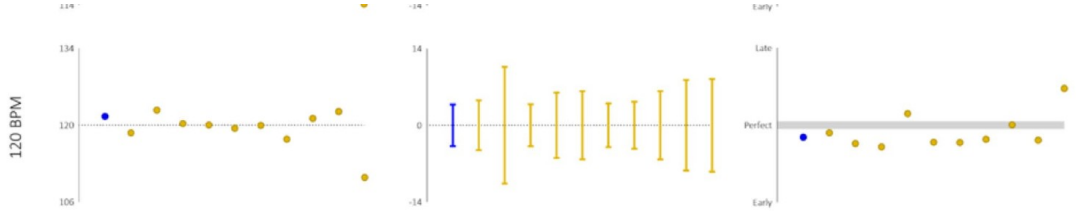
Cook et al. (2025), Fig. 3, 112 bpm row. Performance with a new slow snare tempo.

Precise timing can include an anticipatory offset

A **negative asynchrony** occurs when the selected movement landmark precedes the sound. In Ronan's 2025 snare tests, the average phase at 112 and 120 bpm was negative, while the average at 128 bpm was slightly positive.

This timing pattern is inconsistent with treating every head bob as a delayed reaction initiated only after each sound occurs. A repeating movement can be scheduled in anticipation of the next event while continuing to receive timing information from the current acoustic sequence.

At the same time, anticipation need not place every movement exactly at zero phase. Coordination can be stable and predictive with a systematic offset. Describing the signed relationship preserves that biological timing pattern instead of reducing the response to a simple success-versus-failure label.



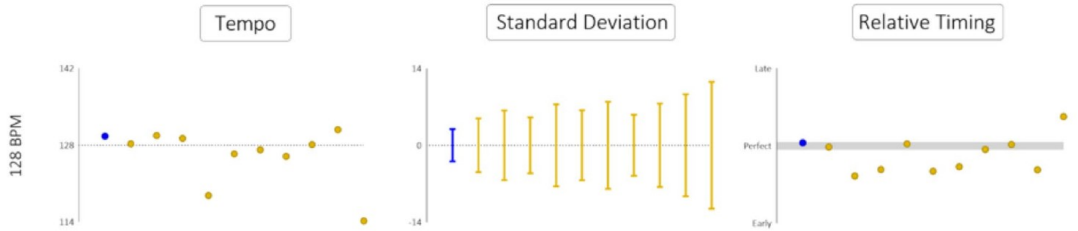
Cook et al. (2025), Fig. 3, 120 bpm row. Timing with the familiar-rate snare sequence.

Rate accuracy and response consistency differ

The human comparison separately evaluated performed tempo, variability of movement intervals, and timing relative to sound onset. These dimensions answer different behavioral questions: how fast the repeated action runs, how consistently it repeats, and where it falls within the external cycle.

Ronan’s intervals were more consistent than those of the human participants across the three test rates. Her average tempo or average phase was also closer to the target than that of many participants, but the relative ranking depended on the performance measure.

No human exceeded Ronan on all measured dimensions. The conclusion rests on matched-task performance rather than a claim that the animal is superior at every possible rhythmic activity. Gross movement, stimulus form, and the selected timing landmark define the behavior that was compared.



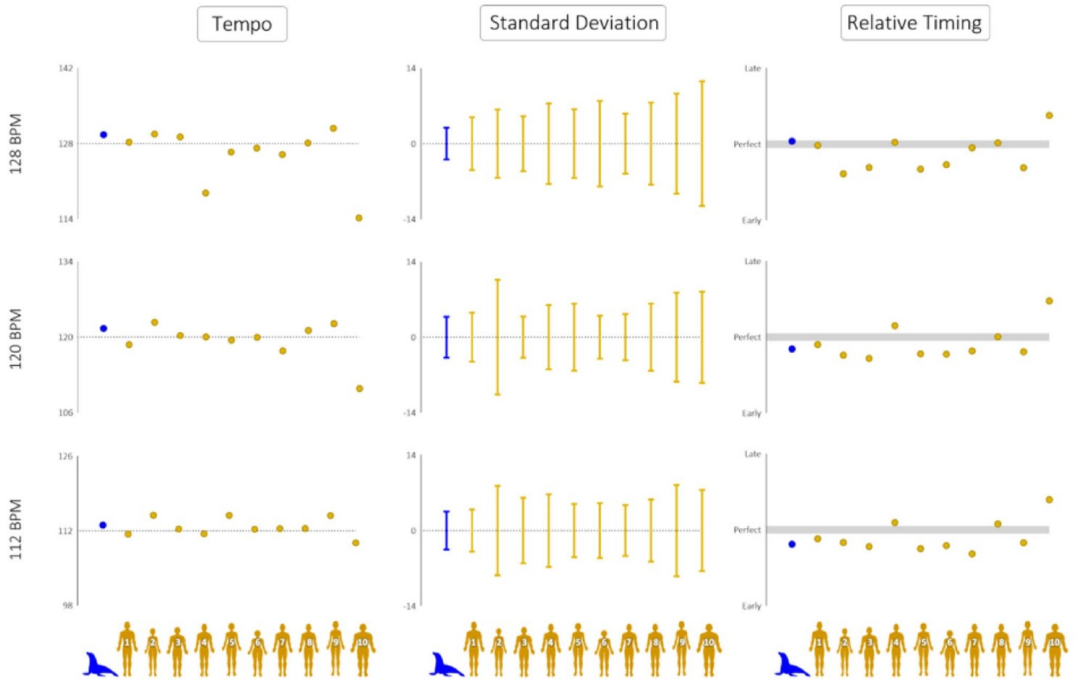
Cook et al. (2025), Fig. 3, 128 bpm row. Different dimensions of synchronized performance.

Human gross movements vary across individuals

Humans in the comparison were non-musicians performing a large arm movement rather than a familiar tapping task. Their synchronization differed among individuals, with some participants producing less consistent movement intervals and more variable phases than others.

The distribution of performance matters because a comparison with only a hypothetical perfectly regular human would conceal the range of actual behavior. The experiment measured the same sound sequences in both species and retained each participant's timing profile.

The study found no consistent rapid improvement across the brief human testing sequence. Ronan, in contrast, entered with a long history of the head-bob task. The observed comparison therefore describes an experienced sea lion and humans performing the specified large-amplitude coordination task, with their different task histories made explicit.



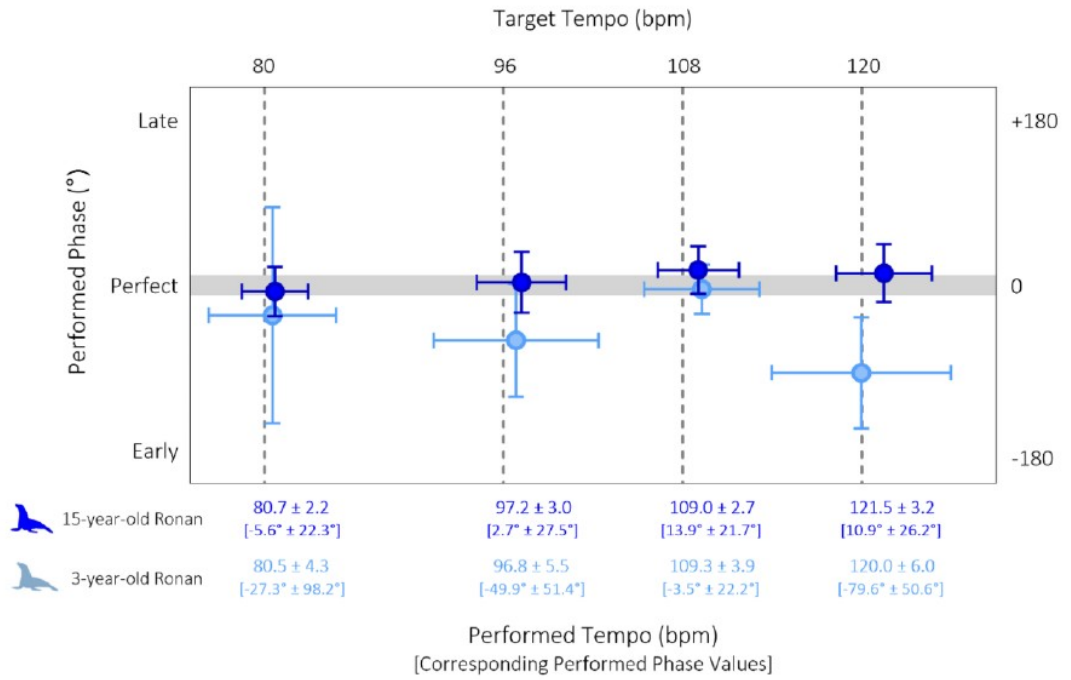
Cook et al. (2025), Fig. 3. Individual human and sea-lion timing measures.

Long-term performance includes improved phase alignment

At the familiar simple-stimulus rates, Ronan's later phase alignment was much closer to zero than in several early conditions. The large negative offsets recorded at slower rates in 2013 were substantially reduced when those stimuli were retested after longitudinal experience.

Improvement involved more than bringing the average movement rate near the stimulus. The timing landmark moved closer to the sound onset, and successive movements became more consistent. Rate, phase, and consistency therefore changed together in the later performance profile.

The 108 bpm condition already had a phase near zero in the early experiment, leaving less room for improvement on that particular measure. This condition-specific pattern connects the long-term result to the actual starting performance rather than assuming identical improvement at every tested rate.



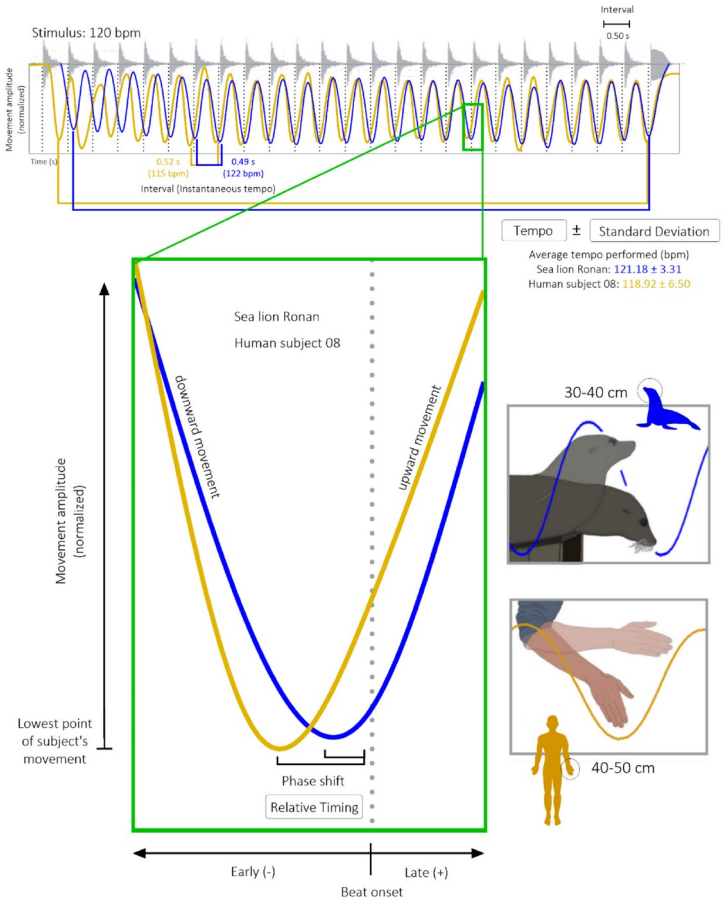
Cook et al. (2025), Fig. 2. Reduced timing offsets in experienced performance.

The tested skill is flexible beat synchronization

Ronan’s experiments established coordination with regular pulse sequences and with musical recordings that contain a regular pulse. **Beat synchronization** is the alignment of a repeated action with those pulse events, with flexibility across rates and acoustic forms.

More complex musical abilities involve additional structure, such as grouping beats into recurring patterns or coordinating with deliberately irregular sequences. The 2025 paper distinguished the measured beat-keeping skill from these broader questions about musical organization.

That distinction changes the scientific interpretation: flexible auditory–motor timing is experimentally supported in this sea lion, while complete human-like musical competence was not the tested outcome. The behavioral result still provides a substantial comparative example of prediction, repeated movement, transfer, and adjustment to external rhythm.



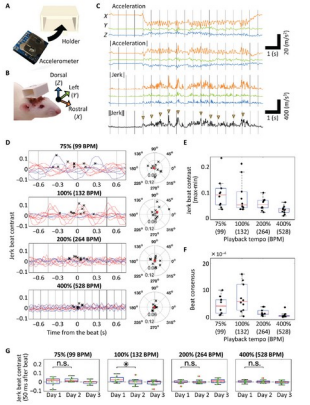
Cook et al. (2025), Fig. 1. Shared movement and timing definitions.

Rats move near musical beats without task training

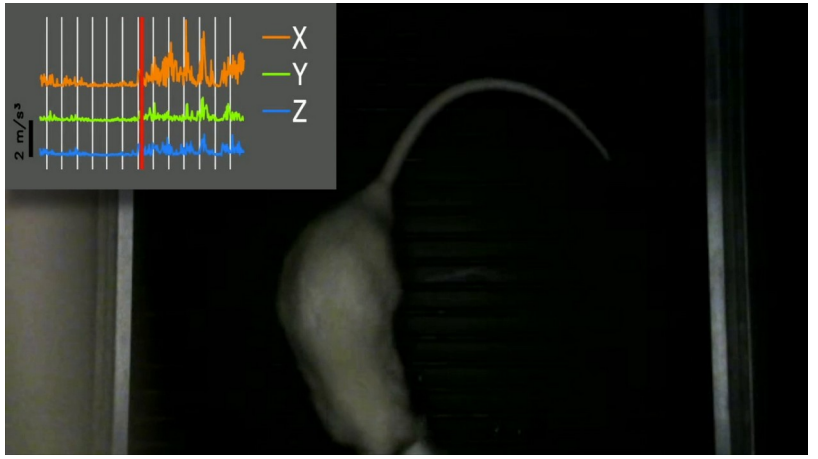
Ito and colleagues attached a lightweight wireless accelerometer to rats and measured small head movements while music played. An **accelerometer** measures acceleration along specified axes, allowing movement changes to be quantified without requiring a trained beat-marking action.

The rats listened to Mozart's Sonata for Two Pianos, K.448, at its original rate and at altered playback speeds. Their movements were more strongly aligned with the musical beats near the original tempo, approximately 132 bpm, than at the very fast versions.

This preparation differed from Ronan's explicitly trained head-bob task. It measured spontaneous small movements during listening rather than an operantly reinforced criterion response. The contrast provides two experimental routes to auditory–motor relationships: acquiring a reliable instructed action and measuring naturally occurring timing during sound exposure.



Ito et al. (2022), Fig. 1A–G. Accelerometer measurements during musical playback.



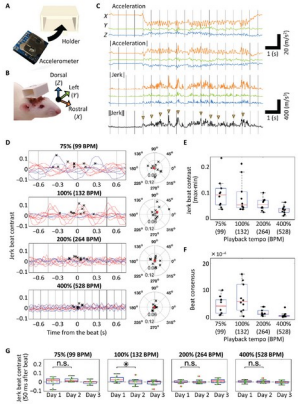
Ito et al. (2022), Movie S1. Rat head movements and simultaneous jerk traces.

Jerk detects rapid changes in head movement

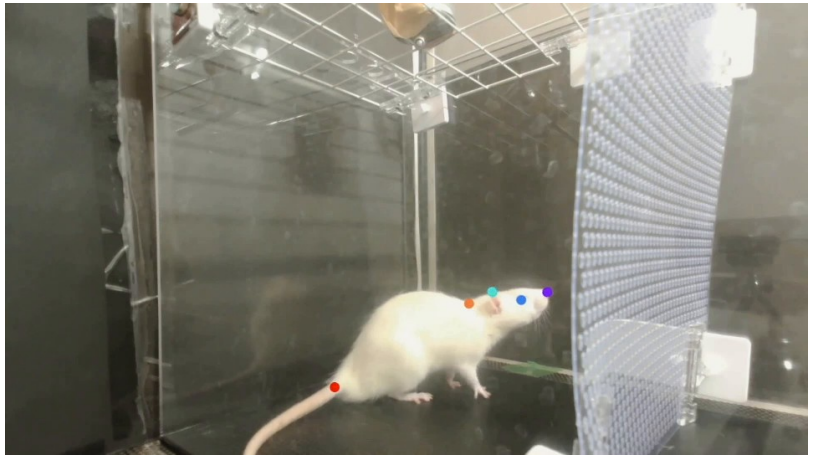
Jerk is the rate of change of acceleration. The rat study used this quantity to identify brief movement changes during musical listening. A small movement with a rapid change in acceleration can therefore contribute strongly to the measured response.

Researchers compared jerk near putative beats with jerk elsewhere in the beat cycle. This beat contrast quantified whether the movement signal was organized around the musical pulse instead of being uniformly distributed through time.

At the original musical tempo, the rats showed beat-related movement contrast. Faster playback weakened that organization. The measured response therefore depended on the sound's temporal scale, connecting the music rate to the timing of small spontaneous movements rather than only to the overall amount of activity.



Ito et al. (2022), Fig. 1C–F. Acceleration, jerk, and beat-relative movement contrast.



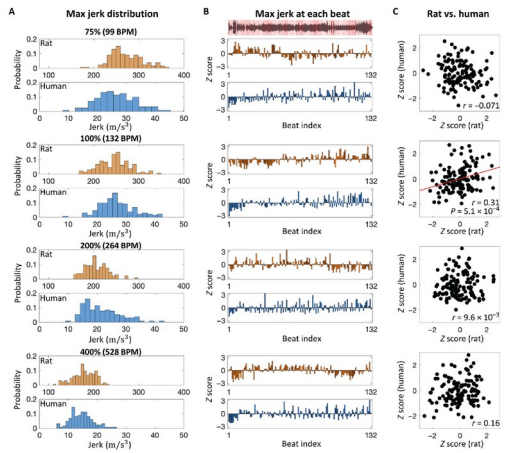
Ito et al. (2022), Movie S2. Beat-related movements in a standing rat.

Rat and human movements favor similar tempos

The experiment also measured head movements in humans listening to the same musical excerpt at altered speeds. This provided a comparison in which both species were exposed to the same changing temporal structure, although their absolute movement magnitudes differed.

Beat-related movement organization was strongest near the original musical rate rather than at the greatly accelerated versions. The two species also showed related changes in movement strength across individual beats of the original excerpt.

Inspection of the stimulus and motion timing indicated that many movements occurred predictively near the upcoming beat. In this preparation, beat-related organization emerged during listening without Ronan's explicit synchronization training, broadening the experimental evidence beyond a single kind of instructed motor task.



Ito et al. (2022), Fig. 2A–C. Beat-related movement comparisons across playback speeds.



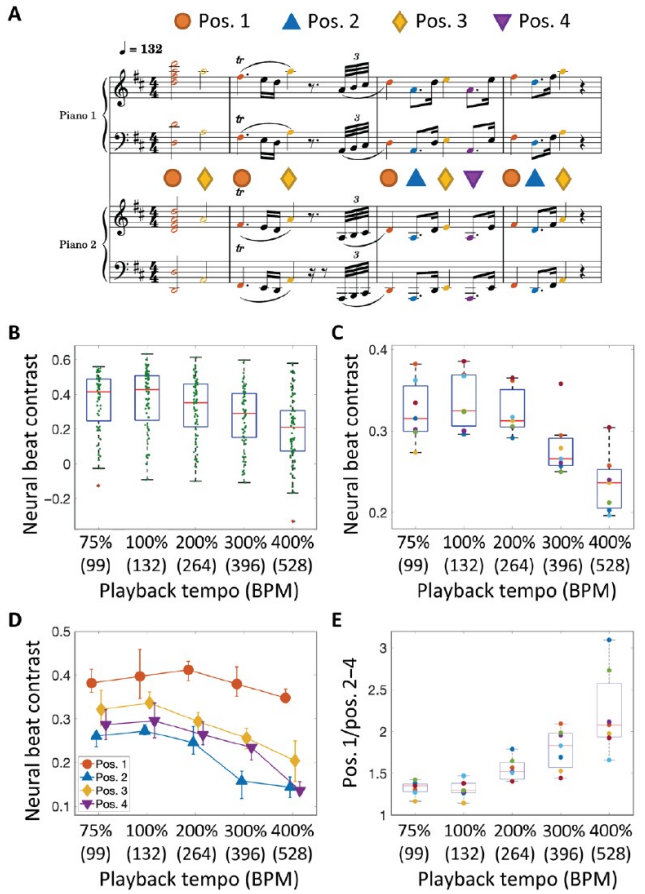
Ito et al. (2022), Movie S3. Human head movements and marker tracking.

Population activity is tuned to musical event timing

The rat study separately recorded **multiunit activity**, the combined spiking response of several nearby neurons, while presenting music at different playback speeds. Activity following beat events was compared with activity following other events in the same musical sequence.

Neural beat contrast was greatest around the original tempo and weakened at much faster speeds. The response also differed among positions in the repeating beat pattern, connecting neural activity to the temporal organization of the musical excerpt rather than merely its presence.

These recordings and the movement measurements were separate experimental assays. Their shared tempo preference linked a candidate neural timing process with the behavioral preference, while the authors used further click stimuli and a model to examine how adaptation could generate that pattern of sensitivity.



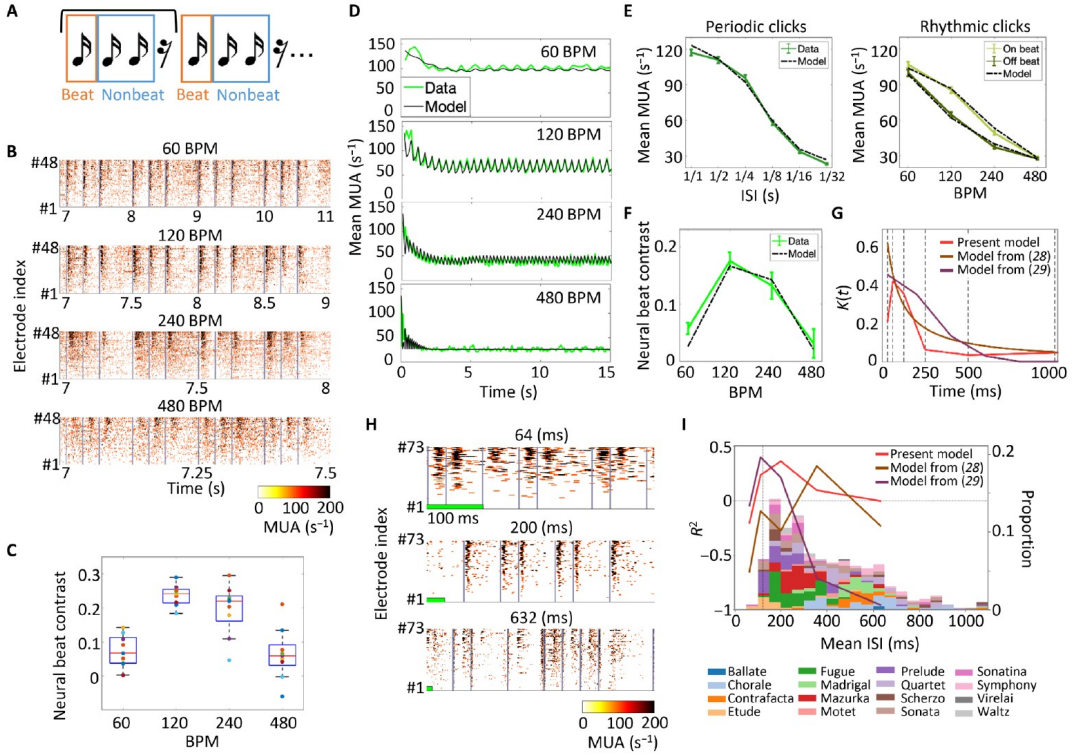
Ito et al. (2022), Fig. 3A–E. Neural responses to musical beats at different rates.

Short-term adaptation favors intermediate rhythms

Short-term adaptation is a change in responsiveness during recent stimulation. Repeated sounds can reduce a neural population's response, with recovery depending on the time since preceding stimulation. This creates a temporal dependence even when each sound is acoustically similar.

Ito and colleagues tested click sequences with three clicks and a rest at 60, 120, 240, and 480 bpm. Recorded responses distinguished beat positions most strongly near an intermediate rate. Very slow sequences and very rapid sequences produced different balances of adaptation and recovery.

The authors' adaptation model reproduced the intermediate-rate preference and related it to the timing structure of musical material. This is a hypothesis about a candidate neural computation; the sea-lion experiments independently established flexible movement correction, rather than directly measuring that rat adaptation process in Ronan.



Ito et al. (2022), Fig. 4A–I. Recorded responses and adaptation-model predictions.

Key takeaways

Tempo and phase A matching repetition rate and a stable relationship to individual beats are distinct aspects of synchronization.

Flexible transfer Ronan applied trained head bobbing to unfamiliar tempos, new sound forms, and musical pulse sequences.

Predictive timing Continued movement through omitted sounds and anticipatory offsets support timing beyond isolated sound-triggered reactions.

Corrective dynamics Phase disturbances change cycle position; persistent tempo disturbances also require adjustment of cycle duration.

Experience and comparison An experienced sea lion performed precisely on a matched gross-movement task, with results depending on the measured timing dimension.

Candidate neural computation Rat recordings and adaptation modeling link intermediate-rate sensitivity to recent sound history; Ronan’s oscillator account describes measured behavioral dynamics.

Cook et al. (2013, 2025); Rouse et al. (2016); Ito et al. (2022)